The Missing Data Bias in Modern Fund Portfolio Data

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Abstract

Despite substantial improvements in the coverage of fund portfolio data in commercial databases since 2010, voluntary reporting remains non-random and heterogeneous, introducing persistent biases. I document key stylized facts about portfolio data missingness in these datasets and introduce a simple decomposition framework that distinguishes absence along three dimensions: reporting participation, frequency, and consistency. Leveraging exhaustive, confidential regulatory data, I find that whilst the reporting frequency is typically uniform within fund families and thus unlikely to introduce major bias, portfolio data is still often unavailable for young funds and fund-months with large flows. Crucially, portfolio characteristics add virtually no explanatory power once these previous factors are accounted for, indicating that strategic non-disclosure plays at most a marginal role in driving portfolio data omissions. I re-examine key results in prominent areas of mutual fund research and show that excluding missing portfolios and non-reporting funds can meaningfully alter empirical conclusions. Finally, I assess three remedies: merging commercial databases, excluding early fund observations, and applying short horizon forward/backward imputation, and evaluate their effectiveness in reducing the missing data bias across contexts.

Keywords: Investment funds, Portfolio disclosure, Missing data

JEL Codes: G11, G23, C80

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1 Introduction

Most research on investment funds uses incomplete commercial databases. Whilst regulation can require funds to report portfolios, this does not automatically lead commonly-used databases to be exhaustive. In some jurisdictions, regulations mandate public disclosure of portfolio holdings[1]. Given that mandatory portfolio disclosure is absent in most regions (e.g., Europe), and that the number of publicly disclosed portfolios is often too low for comprehensive statistical analysis, researchers typically rely on voluntarily reported portfolios gathered by third-party commercial data providers. As of 2024, two of the most popular sources for this data are Morningstar and LSEG Lipper, with LSEG Lipper supplying portfolio data to the CRSP database.

Gaps in these databases are numerous. In the CRSP database, nearly all US funds have at least one missing portfolio. In Lipper, around a third of European funds have no available portfolio at all. Unfortunately, understanding the rationale behind fund portfolio (non-)reporting requires insights into the portfolios of non-reporters to uncover the underlying motivating factors. The research on this question has mainly focused on *how often* funds report a portfolio, ignoring temporary gaps in reporting and permanent non-reporting. The identification of drivers of voluntary portfolio disclosure naturally raises the question of the robustness of empirical findings to the non-randomness of the missing data process. In the fund literature, however, few papers undertake robustness checks to correct for sample selection bias, and there is no prevailing approach. The role of missing data in fund portfolio data and the implications for sample selection bias remains a largely unanswered question.

This paper studies this widespread yet little-researched phenomenon by using exhaustive, confidential regulatory portfolio data, and provides four main contributions: (i) it provides a comprehensive analysis of missing US and European fund portfolio data, (ii) it investigates factors of data absence by studying the drivers of fund portfolio reporting, (iii) it examines the impact of this missingness in a variety of applications, and (iv) it evaluates the effectiveness of different methods for avoiding and correcting the resulting biases.

I start by providing an overview of the institutional details behind portfolio data availability and provide estimates of the types of portfolio data absence in Lipper (CRSP for the US) and Morningstar. I propose a simple approach to distinguish between possible causes of missing data, along three dimensions: the *extensive margin* (whether fund reports their portfolio at all or not), the *intensive margin* (the frequency at which they report) and the consistency of reporting or the *dynamic margin* (whether there are "gaps" or holes in the delivery schedule, short-term or long-term). I find that for US funds in CRSP from 2010/01 to 2024/12, when two thirds of fund-months have an available portfolio, 67% of the missing data can be explained by the intensive margin, 17% by

[1] In the United States, for instance, funds are required to report their holdings to the regulator, which then releases this information to the public at a set frequency. Exchange-traded funds represent the most stringent case, with mandates to report portfolio data daily.

the long-term gaps, and 14% by the short-term dynamic margin. The dynamic margin (short- and long-term) is particularly important between 2014 and 2017, and diminishes after 2018. For European funds in Lipper and Morningstar, where there is no mandatory portfolio disclosure, only around half of fund-months have a provided portfolio, and the share of the extensive margin and long-term gaps in the missing data is significantly higher (a combined 70%).

Next, I investigate the factors that influence the availability of portfolio data. I exploit confidential, monthly Banque de France filings for the full universe of French funds, where public disclosure is voluntary but regulatory reporting is universal, to study the drivers of the three reporting margins. At the intensive margin, fund company–time fixed effects explain the vast majority of variation in reporting frequency, underscoring the dominance of company-level policies. By contrast, the extensive margin is driven by fund-level traits: larger, older funds and those in more competitive markets are significantly more likely to report. The dynamic margin reveals greater temporal heterogeneity: older funds incur fewer short- and long-term gaps, and fund fixed effects alone account for up to half of the residual variation after conditioning on observables—pointing to idiosyncratic, event-driven disclosures. The minimal uplift in explanatory power from portfolio concentration, turnover, and returns suggests that strategic non-reporting by funds is, at best, marginal. Importantly, these patterns indicate that missing portfolio disclosures are systematically non-random, with the potential to bias empirical analyses that rely on observed holdings. Replicating these analyses on other European funds as well as U.S. mutual funds confirms the robustness and generalizability of these findings.

Next, I explore the implications of non-random missingness for empirical fund research by quantifying how each reporting margin distorts performance and behavior measures. I first employ simple fund-month fixed-effect regressions to compare asset flows and returns across observations missing due to long-term and dynamic gaps versus those missing only at the intensive margin. I find that months skipped because of dynamic or long-term reporting gaps display markedly different flow patterns, which is consistent with the drivers of fund reporting established above. Second, I replicate landmark studies of (i) managerial skill, (ii) fund liquidity management, and (iii) socially responsible investment using both the complete regulatory dataset and its commercially available subset. Estimates of manager skill and liquidity-management policy diverge significantly when dynamic gaps are included—often reversing inferences—whereas ESG-fund analyses prove robust to missingness, suggesting minimal strategic concealment. Together, these results underscore that empirical conclusions based on commercial holdings data can be substantially biased if the non-random nature of missing portfolio disclosures is ignored.

Finally, I evaluate three strategies to mitigate missing data challenges. First, merging Lipper/CRSP with Morningstar substantially boosts coverage: for U.S. funds, this merge largely eliminates the dynamic margin of missingness and thereby the missing data bias, whereas for European funds it still leaves pervasive long-term gaps. Second, applying a simple fund filter—dropping each fund's

first six months of data—removes just 3.5 percent of observations yet markedly reduces the differences in flows and returns between missing and observed months. It is therefore a relatively blunt but effective method for mitigating biases introduced by missing portfolio data. Third, I assess imputation: basic forward/backward fills provide a relatively good fit for short-term (one- to two-month) gaps (though performance deteriorates for younger funds), while more advanced, return-driven techniques yield only marginal gains over the simple approach on these horizons. Together, these methods offer practical, context-dependent remedies for non-random missing portfolio data.

Related Literature. The paper contributes to three distinct strands of literature. First, it relates to the literature on voluntary portfolio disclosure[2]. Parida (2019) finds a negative relationship between competition in an investment category and the rate of voluntary portfolio disclosure in that category, a relationship that is amplified for funds holding illiquid assets. George and Hwang (2023) develops a model to explain why mutual funds may benefit from voluntary portfolio disclosure. The greater is the fund's liquidity provision role, the more aggressively the fund trades on its ideas, the stronger is its preference to disclose information about its holdings voluntarily, and the weaker is its performance. Friedman and Mahieux (2024) develops a model to examine how portfolio disclosure affects monitoring spillovers between competing asset managers. Kaniel et al. (2023) studies the effect on performance of quarterly partial disclosure of portfolios in the context of mandatory semi-annual disclosure, finding higher returns in partially-disclosed portfolios. Gervais and Strobl (2020) develops a model of delegated portfolio management in which money managers signal their investment skills via fund transparency. High-skill managers are opaque and rely on returns to be attractive to investors, medium-skill managers instead rely on transparency, and low-skill managers mimic high-skill managers' opacity. Li et al. (2023) studies the determinants of voluntary portfolio disclosure for US equity funds, finding evidence of Gervais and Strobl (2020)'s predictions and finding that funds with higher turnover, higher expense ratios, and higher likelihood of committing fraud tend to disclose their holdings less frequently. Schwarz and Potter (2016) studies voluntary portfolio disclosure by comparing quarterly SEC filings with portfolios provided to commercial databases. In contrast with Li et al. (2023), they find that voluntary disclosure is likely driven by convenience of asset managers rather than duplicity. Investment companies likely find it more efficient to simply report all of their funds' portfolios at calendar quarter-ends when they also report other data (such as 13F filings). A few papers extended this analysis outside of the US, where the absence of additional mandated portfolio disclosure could lead to stronger incentives for strategic usage of portfolio disclosure. Gregory-Allen et al. (2019) and Gimeno et al. (2021) focus on the effects of disclosure on fund returns with 1,344 Australian and New Zealand funds and a set of 118 Spanish equity funds respectively. Chen et al. (2017) studies whether copycat funds could prohibit alpha-generation of funds reporting at higher frequencies with a set of 58 Australian active equity fund managers. To the best of my knowledge, no paper has attempted

[2]It further relates to the literature on the effects of changes in mandatory portfolio disclosure (Agarwal et al., 2014; Choi, 2022).

to test a larger set of drivers of voluntary portfolio disclosure beyond the reporting frequency. I contribute by separately studying multiple dimensions of fund portfolio reporting and identifying gaps stemming from reporting inconsistency as the principal source of missing data bias, comparing portfolio reporting in Europe to the US, and with exhaustive regulatory data test portfolio-level drivers.

Second, this paper contributes to the literature on the potentially biasing effect of missing fund data. Early papers instead addressed how missing fund returns could affect the study of mutual fund performance (Elton et al., 1996, 2001) and whether past empirical analyses remain valid when studied with higher frequency portfolio data (Elton et al., 2010, 2011). Huang et al. (2023) shows that significantly more institutional assets need to be included when estimating the scale of actively managed portfolios, findings that the dollar value added of active strategies is more substantial and persistent than previously estimated. Regarding hedge funds, Joenväärä et al. (2021) and Barth et al. (2023) combine several commercial databases, and discover that average returns are lower and more persistent and that non-reporting funds have less fragile capital and higher alphas. The paper most related to this one is Schwarz and Potter (2016), which tests prominent empirical analyses to compare estimates resulting from commercial and public data, and finds few differences. However, this paper focuses on American equity funds, and uses data up to 2008, for a total of 77,555 portfolios. Therefore, it covers either only a small subset or no portfolio data used in more recent research. For comparison: CRSP provides over 15 times more US fund portfolios from January 2010 to December 2024. Over the same period, Lipper provides nearly 2,5 million portfolios for European funds. This paper diverges in two other critical ways. First, I utilize exhaustive, confidential monthly regulatory data on French funds. Since the monthly frequency reflects the highest frequency of reporting observed among mutual funds, it precludes the commercial dataset from offering unique insights, as it does not contain additional fund portfolios beyond those in my dataset. Second, my methodology extends beyond replication by including a detailed decomposition of portfolio missingness. This approach allows me to identify and test which components of fund portfolio data missingness contribute to variations in results compared to existing empirical analyses. I find that omitting missing portfolios and non-reporting funds can lead to significantly different conclusions from those obtained when all data is considered.

Finally, this paper contributes to the literature on reproducibility and data quality in empirical finance. Recent work underscores the importance of replication and computational reproducibility in economics and finance, emphasizing that robust empirical findings require careful attention to data construction and documentation (Pérignon et al., 2024; Gertler et al., 2018; Herbert et al., 2023). Within this context, a growing strand of research focuses on missing data, particularly in asset pricing, where multiple imputation techniques have been tested and proposed to address gaps in the prominent datasets (Freyberger et al., 2024; Bryzgalova et al., 2024; Chen and McCoy, 2024). In mutual fund research, Farrell (2018), and more recently Bellina et al. (2023) and Bongaerts et al. (2024), develop methods to infer daily portfolio holdings from quarterly filings, outperforming

naive approaches. This paper extends this line of work by documenting patterns of portfolio data missingness in widely used mutual fund databases and adapting these holdings inference techniques to construct monthly portfolio imputations.

The rest of the paper is organized as follows. In section 2, I describe how fund portfolio data is collected by commercial and regulatory databases and document stylized facts of portfolio missingness in CRSP/Lipper and Morningstar. Section 3 investigates the factors that influence the availability of portfolio data. In section 4, I assess the consistency of established empirical results when missing data is considered. Section 5 discusses the effectiveness of different remedies to the missing fund portfolio data bias. Section 6 concludes.

2 Institutional details and Missing data in Commercial databases

2.1 Portfolio Data Collection

Given the dynamic nature of investment portfolios, it is challenging to capture all characteristics that are of interest to both academics and investors using a standard set of metrics. Portfolio data, typically provided as month-end snapshots, offer insights into an investing entity's strategy and can be used to benchmark funds. Commercial data providers (e.g., Lipper) aggregate this information and make it available on their platforms (e.g., Lipper IM), which are in turn used by investment analysts and researchers.

When portfolio data is voluntarily provided, the process of data collection from a fund can be initiated by either the data provider or the fund management company. In the first case, the data provider reaches out to the fund management company to request access to the fund's portfolio data for inclusion on the provider's platforms. The fund management company can respond in one of three ways: it can agree to share the data at a monthly frequency, refuse outright, or agree to share at a frequency lower than the monthly standard preferred by the data provider. In the second case, the fund management company typically responds to an investor request to make the portfolio of a specific fund available. This is more commonly an ad hoc fund-by-fund process.

During portfolio data collection, the fund management company and the data provider agree to an embargo period, during which the data is not yet available on the platform. This period can last from a few days to a few months and is chosen to reduce the perceived risk of rival funds copying the fund's strategy or react strategically to them ("copy-cat funds"). The selection of a lower frequency of reporting is sometimes chosen to the same end.

Regulatory data follows a different procedure. In the US, fund companies are required to send filings to their regulator that include their funds' portfolios at month-ends[3]. The regulator makes certain

[3]Every month-end since the introduction of N-PORT filings in 2018-2020, and the last month-end of every fiscal quarter prior to that.

of these portfolios available to the public on its EDGAR platform. In jurisdictions without this mandatory portfolio disclosure, funds can still engage in portfolio reporting to regulatory institutions. In France, funds are not required to publicly disclose their portfolios. Specifically, according to Article D. 214-31-2 and D. 214-33 of the French Monetary and Financial Code, French investment funds must only report their portfolios to their investors on a half-yearly basis, and disclose this information within four months of the end of the financial period. Yet, French funds report their monthly portfolios to the Banque de France since 2011. This form of confidential disclosure is common in countries without mandatory portfolio disclosure (e.g., Luxembourg, Germany).

In this paper, I study the coverage of commercial databases commonly used in the empirical literature: Morningstar, Lipper and CRSP. With the use of confidential and exhaustive regulatory fund data from the Banque de France, I can thus observe the portfolios that funds have opted not to disclose voluntarily. For clarity, the analyses begin in January 2010, the first period with comprehensive data coverage[4]. Since the OPC data is available from July 2011 onwards, portfolio-level analyses will commence from that point.

2.2 Missing Portfolio Data

2.2.1 Stylized Facts on Missing Portfolio Data

I visualize the availability of portfolio data in the US and Europe. Figures 1 and 2 show the missing (and non-missing) portfolio observations for a random sample of 600 funds from each database[5] in the form of a table, with each row representing one fund and each column one month. A fund-month is green when a portfolio is available in the dataset, red when no portfolio is available, and grey when the fund is not active during that month (either because it is not yet launched, or it is no longer active). This visualization provides a survivorship bias-free view of the availability of portfolio data in the studied dataset. Appendix A provides additional visualizations of the portfolio data coverage in these datasets.

It is immediately obvious from Figures 1 and 2 that the coverage of CRSP and Morningstar in the US is better than their equivalents in Europe, with significantly better reporting especially from 2011 onwards. A further inspection reveals certain distinct patterns in portfolio availability. **First**, the distinct green grid and long green lines exposes the popularity of monthly and quarterly portfolio reporting schedules, especially in the US. Few funds switch between these two regimes, with relatively rare changes appearing to be permanent. **Second**, deviations from an otherwise consistent monthly or quarterly reporting schedule are frequent, with most funds having short-term gaps in their portfolio reporting. Perfectly consistent reporting is very rare, especially among monthly reporting funds. **Third**, both in CRSP and in Lipper, portfolio data for newly established funds

[4]Changing the exact starting date has minimal impact on the visualizations and descriptive statistics, with the exception of CRSP for US funds as coverage stabilizes from December 2010 onwards.

[5]This random sample is limited to 3 to 5% of funds for clarity. The selection of the subsample does not change the visibility of the patterns of missingness.

is frequently unavailable during the initial months following the fund's launch. This observation is not explained by Zhu (2020)'s findings regarding the absence of young and small funds in the merged CRSP-Thomson Reuters database, and are instead a pattern in the CRSP portfolio data (i.e., one of Zhu (2020)'s proposed solutions to the missing small and young funds). This trend is particularly evident between 2012 and 2018, although it remains noticeable in 2024. This is not a form of fund incubation, since my analysis begins at the fund's launch[6]. A comparable pattern preceding a fund's inactivity is observable, although less pronounced. **Fourth**, many European funds do not report portfolios at all. This cannot be attributed to the previous pattern of unavailability of data for recently-launched funds, as some do not report for over a decade.

[6]Empirical studies often exclude incubated funds by discarding all fund months before launch, even though those funds may already report returns and in some cases full portfolios during this period.

Figure 1: Available portfolio data on US funds in CRSP and European funds in Lipper (2010/01-2024/12)

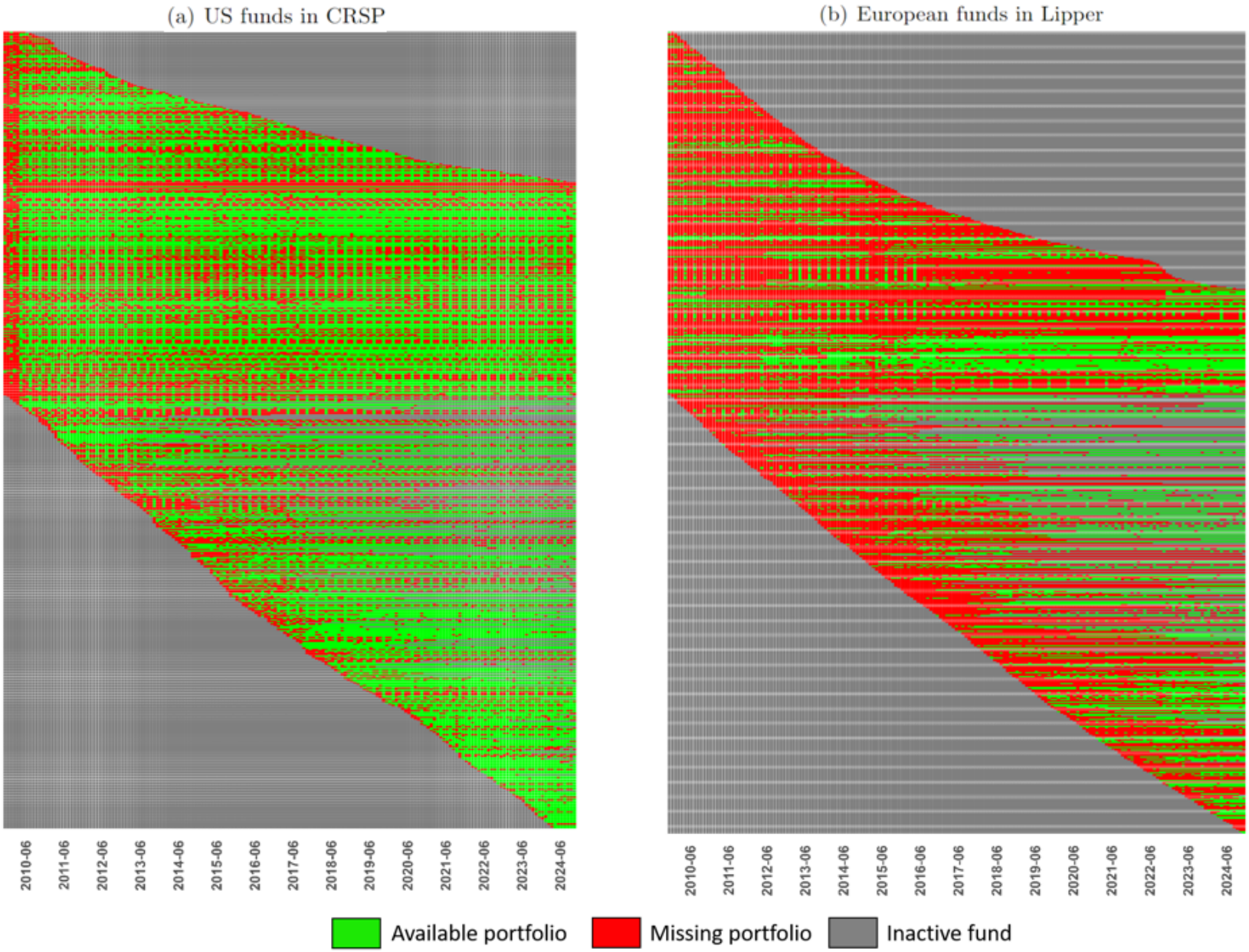


Figure 2: Available portfolio data on US and European funds in Morningstar (2010/01-2024/12)

2.2.2 Decomposition of Missingness

I seek to explore the mechanics of these missingness patterns. To this end, based on the earlier description of the data collection process, I propose a categorization of portfolio data missingness along three dimensions to distinguish their causes and ultimate effects on the representativeness of available datasets. At the ***intensive margin***, a fund can choose between different frequencies at which it reports its portfolio. The most common frequencies are biannual, quarterly and monthly, with regulation in certain jurisdictions imposing a minimum frequency. This is the focus of most of the literature, which exploits US data. At the ***extensive margin***, a fund can decide to never report a portfolio. In this case, no contemporaneous information on investment strategies, risk and exposures can easily be inferred. The extensive margin is of greater interest in jurisdictions without mandatory minimum reporting frequencies, as is the case in Europe, eastern Asia, Australia and New Zealand. A fund can further choose to engage in ***dynamic reporting*** - reporting with a (mostly) consistent frequency, but with gaps. These missing months can be attributed to random factors such as data pipelines of the fund's parent company, as well strategic non-reporting during period of heightened information sensitiveness. These first three dimensions are the focus of this paper[7].

Identifying the extensive margin is straightforward, as it consists of identifying funds without recorded portfolio. To identify the intensive margin, I utilize the local mode of reporting. Specifically, if a fund consistently reports a portfolio three months after the previous one, it is classified as a quarterly-reporting fund. To account for instances where a portfolio might occasionally be missing, I smooth the reporting frequency by examining the local mode across the three preceding and three subsequent reported portfolios. This approach ensures that the classification of the reporting frequency is robust to occasional lapses in data reporting, providing a more accurate representation of the fund's typical reporting behavior. Additionally, this method is flexible enough to capture changes in reporting patterns, such as a transition from quarterly to monthly reporting. Overall, this approach leads to a classification of 99.1% of US and 98.8% of European fund-months from 2010 to 2024 to either non-reporting or reporting at a either a monthly, quarterly or semi-annual frequency[8]. This then allows for the distinction of fund-months that are missing because of dynamic reporting, which is the remainder. Inactive funds naturally do not report portfolios, so this categorization and resulting estimates focus on the period between a fund's launch and its closure [9].

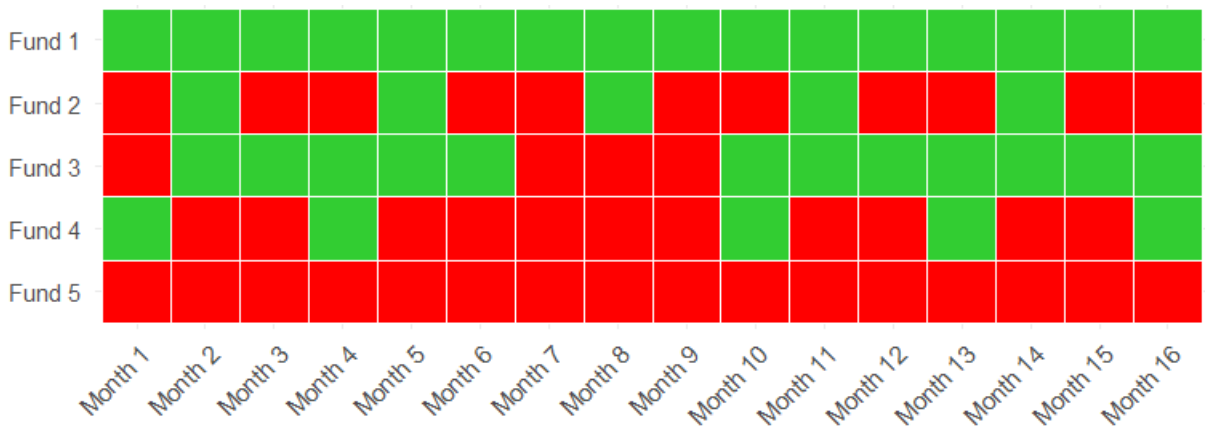
[7]The ***completeness*** of reporting - how representative a provided portfolio data is of the actual positions of a fund - could be seen as a fourth dimension. In principle, it could be beneficial (although risky) for a fund to leave out certain positions held at a given date (e.g., controversial assets that are at odds with the fund's objective). This does not appear to be a concern in the literature, and for individual funds would be unverifiable. It is worth mentioning, however, that there is often no certainty as to the validity of the portfolio data. In the US for example, the SEC's website EDGAR states that it does not correct errors in regulatory filings. To alleviate this risk of errors, commercial data providers typically perform additional verifications, (e.g., plausability of a fund's portfolio given its returns).

[8]The remainder are mostly cases of long-term non-reporting.

[9]To avoid erroneously identifying a fund as inactive based on the launch and closure dates of individual share classes, I take the earliest launch data and the latest closure date among a fund's share classes. If a fund discontinues

Figure 2 provides the reporting schedule of 5 example funds over a 16 month period. The first row is a consistent monthly-reporting fund, the second a consistent quarterly-reporting fund, the third an inconsistent monthly-reporting (a monthly-reporting fund with anomalies, i.e., dynamically reporting) fund, the fourth an inconsistent quarterly-reporting fund, and finally the fifth a non-reporting fund. Of the 80 fund-months, 21 are therefore missing because of the intensive margin (the second and fourth funds), 5 because of dynamic reporting (the third and fourth funds), and 16 are missing because of the extensive margin (the fifth fund).

Figure 3: Types of portfolio reporting behaviors



Note: Each row identifies a fund, each column a month in which a fund did (green) or did not (red) report its portfolio.

The breakdown of missingness introduced above is helpful to understand the limits of even the most complete portfolio data databases, and compare databases among one another. Table 1 provides the decomposition of missingness for the two studied databases. In CRSP, over the period 2010/01 to 2024/12, portfolio data covers 65% of fund-months, and two thirds of the missing portfolios can be attributed to the chosen reporting frequency. Over the same period, portfolio data in Lipper covers 41% of European fund-months, but the intensive margin only represents 13% of the missing data. I distinguish between short-term gaps in reporting from longer ones, which are here 1) gaps without prior or subsequent portfolios available (e.g., missing portfolios for newly-created funds) 2) gaps lasting at least 12 months[10]. Most can be attributed to long-term missing data and the extensive margin.

The distribution of missing data varies across fund asset types and time. In CRSP, data on US mutual funds is more often missing for domestic than foreign equity funds, and more for corporate than government bond funds. The share of dynamic and long-term missing months reaches on average between 12% and 17% over the period 2014/01-2016/12, and decreases significantly after

its oldest share, a fund is mechanically declared younger than it truly is. To avoid this survivorship bias, I therefore extend this approach to all historical share classes of a fund, both active and inactive.

[10]This 12 months cut-off is informed by looking at the distribution of lengths of reporting gaps, and allows a semi-annual reporting fund to omit one portfolio and still be categorized as a short-term gap. The effect of selecting a different cut-off is marginal.

the introduction of N-PORT filings. In Lipper, there is a more limited but continuous improvement of data quality of European mutual funds since 2016. A breakdown of missing portfolios through time in the US (CRSP and Morningstar) and Europe (Lipper and Morningstar) is provided in Appendix A.

Figure 5 in Appendix A informs on the evolution of these drivers through time by displaying the reporting frequency chosen by funds. Over the past 15 years, the share of monthly-reporting funds has increased substantially, most notably in Europe. In the US and in Europe, over half of the funds report at a monthly frequency ($\frac{2}{3}$ of Total Net Assets (TNA)) in 2024, up from 42% and 33% in 2011 ($\frac{1}{3}$ of TNA). Monthly reporting surpassed 50% of total TNA in the US in 2018 and as early as 2013 in Europe. Yet, the sheer number of non-reporting funds and funds with long-term gaps in reporting can raise concerns in studies on fund-level dynamics even in most recent years. These differences of data coverage and types of missingness raise the question of whether identical analyses performed on US and European data could yield different results due to sample selection.

Table 1: Portfolio Data and categorization of portfolio data missingness in CRSP and Lipper

(a) CRSP (US), 2010/01 to 2024/12

Availability/missingness	**Observations**	**Share of fund-months (%)**	**Share of Missing (%)**
Portfolio available	1,173,701	64.87	-
Missing:	635,572	35.13	100
Intensive	423,758	23.42	66.67
Extensive	17,247	0.95	2.71
Dynamic (short-term)	88,492	4.89	13.92
Dynamic (long-term)	106,075	5.86	16.69

(a) Morningstar (US), 2010/01 to 2024/12[11]

Availability/missingness	**Observations**	**Share of fund-months (%)**	**Share of Missing (%)**
Portfolio available	1,123,677	72.99	-
Missing:	415,802	27.01	100
Intensive	370,053	24.04	89.00
Extensive	562	0.04	0.14
Dynamic (short-term)	14,933	0.97	3.59
Dynamic (long-term)	30,254	1,97	7.28

[11]The values for U.S. funds from Morningstar are based only on those Morningstar funds that could be matched with funds in CRSP. Because this matching was incomplete, the total number of fund-months is lower. In contrast, for European funds, all Morningstar and Lipper funds could be successfully matched, so this limitation does not apply.

(b) Lipper (Europe), 2010/01 to 2024/12

Availability/missingness	**Observations**	**Share of fund-months (%)**	**Share of Missing (%)**
Portfolio available	2,448,451	38.77	-
Missing:	3,867,017	61.23	100
Intensive	382,742	6.06	9.90
Extensive	1,244,636	19.71	32.19
Dynamic (short-term)	507,803	8.04	13.13
Dynamic (long-term)	1,731,836	27.42	44.78

(b) Morningstar (Europe), 2010/01 to 2024/12

Availability/missingness	**Observations**	**Share of fund-months (%)**	**Share of Missing (%)**
Portfolio available	3,616,949	57.26	-
Missing:	2,699,368	42.74	100
Intensive	431,703	6.83	15.99
Extensive	1,088,135	17.23	40.31
Dynamic (short-term)	226,837	3.59	8.40
Dynamic (long-term)	952,693	15.08	35.29

3 Causes of Missingness: Determinants of Disclosure

Portfolio data availability varies across time and across regions, which prompts an investigation into its underlying causes. As mentioned earlier, a first answer lies in regulatory differences[12], but does not fully explain the observed variations. In the US, over half of the reporting funds are reporting at a monthly frequency. In Europe, there is no mandatory public disclosure, and yet many funds opt to disclose their portfolios, most of which at a monthly frequency. What differentiates reporting from non-reporting funds, and what factors influence their reporting frequency and consistency? This section addresses these questions.

3.1 Empirical Strategy

My empirical strategy leverages the confidential regulatory French OPC data from the Banque de France, which offers three key advantages. First, most importantly, it permits a comprehensive analysis of data missingness patterns without risk for major missing data biases. A common problem in studying data missingness is the impossibility to test on unavailable data. On funds, the ideal test of the choice of reporting requires the complete set of portfolios, including those not reported. This is possible by matching reported portfolios with the complete OPC dataset. Second, the high frequency of reported portfolios in OPC facilitates a multidimensional study of portfolio reporting. The data is reported at month-ends, which matches the highest frequency available in commercial databases. It thus becomes possible to study the drivers and effects of the choice of reporting and the choice of reporting at specific frequencies. Third, the data coverage of France is particularly valuable to understand pervasive attributes of fund portfolio missingness patterns. France hosts a diverse array of investment funds and ranks among the world's largest domiciles for fund assets under management. Additionally, the proportion of missing data in France exceeds the European average. A study on the fund disclosure behaviors of French funds are therefore likely to inform on behaviors of funds outside of France.

To identify the determinants of fund portfolio reporting, I run panel logit regressions with a dummy of reporting as the dependent variable. I tackle the incidental parameter bias problem by applying the analytical bias correction derived by Fernández-Val and Weidner (2016). As regressors, I include fund characteristics that can be observed ($ObservableTrait_{j,f,t}$) and a set of portfolio-level variables that cannot be observed ($PortfolioTrait_{i,f,t}$) without portfolio data. If neither fund nor portfolio characteristics are significant, this signals that the data is completely missing at random (MCAR), with implies that the estimates of analyses using fund portfolio data are unlikely to be biased. If I find that none of the portfolio-level variables are significant predictors after controlling for the fund management company and observable fund-level characteristics, that would suggest that the data is missing at random (MAR). Otherwise, it would be a signal that portfolio data is not missing at random (NMAR). In these last two cases, appropriate methods need to be employed to avoid

[12] In the US, mandatory N-SCR and N-Q filings ensured that portfolio data would be publicly available at a minimum frequency.

biases.

I run these regressions in two sets. First, I test the determinants of the **extensive** and **intensive margins**. As these reporting policies change either not at all or rarely, I use the following specification once for the extensive and once for the intensive margin:

$$\ln\left(\frac{P(\text{Reporting}_{f,t}=1)}{1-P(\text{Reporting}_{f,t}=1)}\right) = \beta * ObservableTrait_{f,t} + \gamma * PortfolioTrait_{f,t} + \alpha_{company,t} \quad (1)$$

with $Reporting_{f,t}$ the dummy identifying whether the fund reports at the specified margin (extensive or intensive), $ObservableTrait_{f,t}$ the observable, time-varying fund characteristics, $PortfolioTrait_{f,t}$ the set of portfolio-level characteristics, and $\alpha_{company,t}$ the fund company-time fixed effect. I restrict the set of funds to equity, fixed income and mixed asset mutual funds known to Lipper[13]. I further drop funds that share the same asset manager with fewer than four other funds to avoid mechanically increasing the effect of fund company fixed effects[14]. Finally, I exclude feeder funds and single-stock funds.

Second, I study the determinants of **dynamic reporting**. Since reporting can vary every month, the specification includes current fund flows and returns, and a fund fixed effect to specification (1):

$$\ln\left(\frac{P(\text{Reporting}_{f,t}=1)}{1-P(\text{Reporting}_{f,t}=1)}\right) = \beta * ObservableTrait_{f,t} + \gamma * PortfolioTrait_{f,t} + \alpha_f + \alpha_{company,t} \quad (2)$$

with $Reporting_{f,t}$ the dummy identifying whether the fund provides a portfolio on a given month, α_f the fund fixed effect, and $\alpha_{company,t}$ the time-varying fund company fixed effect. With this fixed effet, I narrow the selection of funds to those mutual funds present in the Lipper IM database that, at some point, failed to report their portfolios for at least one month.

I select observable fund characteristics based on the current literature. First, I include the fund's age (in years) and its size (log of millions of euros). Second, in line with the predictions of Gervais and Strobl (2020) that high-skilled and low-skilled funds are less transparent than medium-skilled funds, I add dummies identifying whether a fund is in the top or bottom quintile in the distribution of five-factor alphas[15] within its asset type-geographical group in a given month[16], Third, following Parida (2019), I add a measure of a fund's competition, which I construct as the log of the number of funds with the same asset class and geographical focus as the fund of interest. However, since a fund's choice to report could vary across dimensions and competitive pressure can translate to a pressure to engage in portfolio transparency, I further add two additional measures of reporting

[13] This drops funds with "nicher" asset classes such as real estate funds, money market funds and alternative funds.

[14] This reduces the sample by a negligible number of funds in France, and by about 600 funds for European funds. Running the regressions without this filter does not substantially affect the estimates.

[15] The factor data taken here to compute a fund's return alpha is the European set, obtained from French's website.

[16] This is identified with the "Geographical focus" column in Lipper IM.

pressure: the share of the fund's competitors that report at all (extensive margin) and that report at a monthly frequency (intensive margin)[17]. I further add as explanatory variable a dummy that identifies whether a fund's financial year is aligned with a calendar year[18], as Schwarz and Potter (2016) finds that this variable is a key driver of voluntary portfolio reporting for equity funds in the US pre-2008.

Next, I base the selection of portfolio-level fund characteristics on economic motivation for fund portfolio opacity. Since a fund's portfolio can reveal managerial skill, which a fund could choose to reveal publicly, I include a measure identified in the literature as a portfolio-level predictor, namely the average monthly portfolio turnover (Pástor et al., 2017). Second, as a fund can choose not to disclose its portfolio to reduce risks of the replicability of its strategy (Verbeek and Wang, 2013), I further add its portfolio's concentration (proxied by HHI).

I replicate these analyses on both European funds (Appendix C1) and US funds (Appendix C2). With the exception of portfolio characteristics, the regressions are identical. The final samples consist of 4,594 French funds, 34,200 European mutual funds, and 17,972 US mutual funds.

3.2 The Extensive, Intensive, and Dynamic Margins

Table 2 reports the estimates for the extensive margin (whether a fund ever reports) and the intensive margin (the frequency of reporting, conditional on reporting), and Table 3 presents the dynamic-margin results (the probability of missing portfolio disclosures over short and long horizons). At the intensive margin, once fund company–time fixed effects are included (Columns 6–7 in Table 2), 81 percent of fund-month observations are dropped, indicating that most asset-management groups enforce a uniform reporting schedule across their funds. This finding implies that virtually all remaining variation in reporting frequency stems from unobserved company-level policies rather than from individual fund characteristics[19]. In contrast, when I model the extensive margin with the same fund company–month fixed effects (which reduces the sample by roughly 20 percent), several fund-level attributes continue to predict whether a fund ever reports. Specifically, larger and older funds and those operating in more competitive markets exhibit a higher propensity to report.

Turning to the dynamic margin, Table 3 reveals that only one-third of fund-month observations display homogeneous reporting behavior within the same management company, underscoring greater temporal heterogeneity than at the intensive margin. Older funds are significantly less prone to both short- and long-term reporting gaps, and a fund's probability of reporting in any given month increases when other funds in its company report, especially those with high reporting frequency.

[17] This share ignores the presence of the fund of interest itself, to avoid creating a mechanical upward bias. As a result, a reporting fund has slightly lower competitive pressure than a non-reporting fund in the same group.

[18] I.e, whether or not it ends on either the end of March, June, September or December.

[19] This is further supported by examples of simultaneous frequency switching by all funds of the same fund company on the same month. For example, in February 2019, the switching of over 200 German funds from quarterly to monthly reporting coincides with the acquisition of their fund company by private equity firm Montagu.

Moreover, for long-term gaps, fund fixed effects alone account for approximately 25 percent of observations and explain about half of the variation remaining after accounting for observable covariates. Across all three margins, I examine whether portfolio characteristics—namely, concentration and turnover—affect reporting decisions. Although higher concentration and lower turnover are statistically significant predictors of lower reporting propensity, their inclusion raises the pseudo-R^2 by at most 0.1 percentage points, indicating that their economic impact is negligible.

I replicate these regressions on (i) European funds (Appendix C1) and (ii) U.S. mutual funds (Appendix C2). The European results closely mirror those for the French sample, with only a slightly attenuated size effect on the extensive margin and a modest positive coefficient on fiscal-year alignment with the calendar year. In the United States (where periodic portfolio disclosures to the SEC are mandatory) I focus on the intensive margin and again find that funds open to new investors voluntarily report more frequently than required, while fiscal-year alignment retains a small but significant effect, in line with Schwarz and Potter (2016).

3.3 Discussion

The foregoing analysis yields two main insights. First, the extensive, intensive, and dynamic margins of portfolio reporting are governed by distinct forces. Reporting frequency is predominantly a company-level decision, whereas the decision to report at all and the occurrence of long-term gaps depend substantially on fund age, size, and competitive pressures. The fact that fund fixed effects explain up to half of the residual variation in long-term gaps points to the importance of unobserved, fund-specific factors, such as idiosyncratic events or explicit investor requests, that trigger reporting months or even years after a fund's launch. Interviews with heads of reporting at European management companies confirm that portfolio disclosures are often released only after a major fund-specific event or upon direct investor demand.

Second, portfolio characteristics, flows, and returns play only a minor role in driving reporting decisions once company-level policies and fund heterogeneity are accounted for. This contrasts with Li et al. (2023), whose findings of strategic non-reporting could potentially reflect their omission of management-company controls. While we cannot rule out the presence of some strategic behavior, these results indicate that such behavior explains little of the overall variation in reporting. Although I find no large-scale evidence of deliberate non-reporting, portfolio data are not missing completely at random, particularly along the dynamic margin. The persistence of reporting gaps suggests potential biases in empirical studies that rely on reported holdings.

Table 2: Drivers of the extensive and intensive margins of voluntary portfolio reporting in France, logit

This table displays the panel logistic regressions on the extensive and intensive margins of reporting for French funds, from 2014 to 2024. Standard errors are clustered at the fund and month-level. I tackle the incidental parameter bias problem by applying the analytical bias correction derived by Fernández-Val and Weidner (2016). 22% of fund-months are of fund companies with homogeneous reporting at the extensive margin, against 81% at the intensive margin. Portfolio-level characteristics are statistically significant but economically insignificant predictors of the extensive margin of reporting.

	Dependent variable:							
	Extensive Reporting				Intensive Reporting			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Log of Fund TNA	0.13***	0.11***	0.14***	0.13***	0.03	0.04	0.02	0.00
	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)	(0.03)	(0.06)	(0.06)
Log of Fund Age (in years)	0.43***	0.36***	0.52***	0.54***	0.18**	0.19**	0.33*	0.36*
	(0.03)	(0.03)	(0.05)	(0.05)	(0.06)	(0.06)	(0.15)	(0.15)
Alligned Reporting	0.50***	0.41**	-0.08	-0.11	-0.81*	-0.76*	-1.22*	-1.27*
	(0.15)	(0.15)	(0.23)	(0.23)	(0.36)	(0.37)	(0.55)	(0.59)
Open Fund	1.58***	1.62***	2.30***	2.31***	-0.34	-0.35	2.42**	2.36**
	(0.20)	(0.20)	(0.28)	(0.29)	(0.48)	(0.47)	(0.85)	(0.86)
Share Institutional	0.28*	0.09	0.17	0.16	-1.62***	-1.47***	1.08	1.13
	(0.13)	(0.14)	(0.19)	(0.19)	(0.28)	(0.29)	(0.65)	(0.63)
Competition (log)		-0.11**	-0.17***	-0.16***		-0.06	0.03	0.03
		(0.04)	(0.05)	(0.05)		(0.06)	(0.10)	(0.10)
Competition Reporting (ext)		4.24***	5.18***	5.15***		-0.01	2.26	1.96
		(0.38)	(0.65)	(0.64)		(0.64)	(1.59)	(1.65)
Competition Reporting (int)		1.11*	1.36*	1.34*		0.30	4.39***	4.36***
		(0.44)	(0.64)	(0.62)		(0.81)	(1.26)	(1.29)
Bottom Quintile Alpha		-0.34***	-0.66***	-0.68***		0.29	-0.53	-0.41
		(0.10)	(0.13)	(0.13)		(0.17)	(0.36)	(0.36)
Top Quintile Alpha		-0.03	-0.06	-0.09		-0.42**	-0.26	-0.29
		(0.11)	(0.14)	(0.14)		(0.16)	(0.33)	(0.34)
HHI				-0.55**				-2.05*
				(0.19)				(0.89)
Turnover				1.30***				-1.02
				(0.27)				(0.81)
Time FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fund Company-time FEs	No	No	Yes	Yes	No	No	Yes	Yes
Observations	268150	267392	208762	208757	96031	95830	17920	17920
Deviance	283774.45	271627.03	161644.45	160313.68	70978.91	70341.65	11122.69	10943.86
Pseudo R^2	0.08	0.11	0.28	0.28	0.05	0.06	0.44	0.45

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table 3: Drivers of the dynamic margin of voluntary portfolio reporting in France, logit

This table displays the panel logistic regressions on the dynamic margin of reporting for French funds, from 2014 to 2024. Standard errors are clustered at the fund and month-level. I tackle the incidental parameter bias problem by applying the analytical bias correction derived by Fernández-Val and Weidner (2016). 37% of fund-months are of fund-companies with homogeneous reporting frequencies at the dynamic margin, and 32% for long-term gaps. Portfolio-level characteristics are not economically significant predictors of reporting.

	Dependent variable:							
	Dynamic Reporting				LT Reporting			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Log of Fund TNA	0.02***	-0.02	0.08**	0.08**	0.07***	-0.10***	0.03	0.02
	(0.00)	(0.02)	(0.03)	(0.03)	(0.00)	(0.02)	(0.04)	(0.04)
Log of Fund Age (in years)	-0.12***	0.11**	0.14*	0.14*	0.17***	2.33***	2.02***	1.94***
	(0.01)	(0.04)	(0.07)	(0.07)	(0.01)	(0.05)	(0.11)	(0.11)
Share Institutional	-0.27***	-0.54***	0.11	0.09	-0.59***	-1.09***	0.76**	0.59*
	(0.03)	(0.09)	(0.18)	(0.18)	(0.02)	(0.11)	(0.28)	(0.28)
Competition (log)	-0.03***	0.46***	0.21	0.22	-0.08***	0.26*	-1.46***	-1.22***
	(0.01)	(0.09)	(0.17)	(0.17)	(0.00)	(0.10)	(0.26)	(0.27)
Competition Reporting (ext)	-0.46***	-0.43	0.81	0.93	1.16***	9.50***	6.28***	7.21***
	(0.09)	(0.48)	(0.89)	(0.89)	(0.06)	(0.58)	(1.35)	(1.38)
Competition Reporting (int)	0.85***	-1.78***	-1.59**	-1.63**	2.05***	5.53***	4.51***	4.15***
	(0.09)	(0.31)	(0.59)	(0.60)	(0.06)	(0.31)	(0.74)	(0.75)
Return	-0.01	-0.00	-0.01	-0.01	-0.01*	-0.00	0.00	0.00
	(0.00)	(0.00)	(0.01)	(0.01)	(0.00)	(0.00)	(0.01)	(0.01)
Flow	-0.00*	-0.00	-0.00*	-0.00*	-0.00***	-0.00	-0.00	-0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
HHI				-0.32*				-1.86***
				(0.16)				(0.24)
Turnover				-0.88***				0.06
				(0.18)				(0.28)
Time FEs	Yes	Yes	NA	NA	Yes	Yes	NA	NA
Fund FEs	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Fund Company-time FEs	No	No	Yes	Yes	No	No	Yes	Yes
Observations	117649	111320	70455	70454	160957	119346	81123	81123
Deviance	120603.10	102356.70	36442.35	36402.69	206931.15	79836.95	18982.82	18880.60
Pseudo R^2	0.075	0.182	0.528	0.529	0.065	0.517	0.821	0.822

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

4 Biases from Incomplete Portfolio Coverage

The non-random nature of portfolio data missingness raises concerns about the robustness of empirical findings in mutual fund research. Schwarz and Potter (2016) explore this with equity fund portfolio data from 1996 to 2008 with the Thomson Reuters database and mandatory SEC filings, finding some differences in estimates but overall no differences in resulting conclusions.

I revisit the issue using post-2010 data and complete confidential regulatory filings. I begin by assessing whether missing portfolio observations are systematically associated with extreme fund-month outcomes, such as abnormal returns or flows. I then examine how these disclosure gaps affect standard empirical results by replicating established analyses using both commercial and regulatory portfolio data. This approach enables a systematic evaluation of the potential inferential biases arising from incomplete coverage in widely used datasets.

4.1 Differences in Returns and Flows by Portfolio Disclosure

In most studies on funds or their behavior, fund returns and/or fund flows form key variable(s) - either as the main dependent or independent variable of interest, or an important control. If fund portfolios are missing in a manner that correlates with either variable, it is probable that it will affect fund analyses.
To determine how randomly portfolio is missing with respect to flows and returns, I run simple regressions with fund and month fixed effects:

$$DependentVariable_{f,t} = \beta * MissingType_{f,t} + \alpha_f + \alpha_t + \epsilon_{f,t} \tag{3}$$

with $DependentVariable_{f,t}$ fund-month returns, flows, next and past quarter returns, and next and past quarter flows, and $MissingType_{f,t}$ the type of missingness as defined in section 2: Long term missing (long gaps in the reporting schedule), dynamic missing (short gaps in the reporting schedule) and intensive missing (resulting from a lower reporting frequency). Values are winsorized at the 1st and 99th percentile. Standard errors are clustered at the fund and month-level. Table 4 summarizes the results.

For both the US and Europe, months without portfolio data because of the intensive margin do not have significantly different returns and flows, which is consistent with the data collection procedure. However, months without portfolio data because of dynamic or long-term non-reporting tend to have significantly higher net investor flows. In the US, the quarter following dynamic non-reporting has gross returns 17,5 bps smaller than other quarters. These results suggest that fund-months without portfolio available in CRSP are missing because of inconsistent gaps in reporting, which could bias fund-level analyses.

Table 4: Differences in returns and flows for fund-months with and without available portfolio data, in the US and Europe, by type of data absence

CRSP for US funds (2010/01 to 2024/12)

	Return	Past Return	Next Return	Flows	Past Flows	Next Flows
Intensive Missing	0.029**	-0.047**	0.007	-0.019	0.376**	-0.068
	(0.014)	(0.022)	(0.021)	(0.075)	(0.156)	(0.154)
Dynamic Missing	0.002	-0.021	-0.093**	0.230***	0.681***	0.629***
	(0.028)	(0.051)	(0.045)	(0.071)	(0.198)	(0.164)
Long-Term Missing	-0.019	-0.021	0.084	3.610***	5.261***	16.031***
	(0.058)	(0.092)	(0.099)	(0.260)	(0.452)	(1.054)
Time FEs	Yes	Yes	Yes	Yes	Yes	Yes
Fund FEs	Yes	Yes	Yes	Yes	Yes	Yes

*p<0.1; **p<0.05; ***p<0.01

Lipper for European funds (2010/01 to 2023/12)

	Return	Past Return	Next Return	Flows	Past Flows	Next Flows
Intensive Missing	-0.051	-0.080	-0.056	-0.162**	-0.207	-0.407**
	(0.048)	(0.074)	(0.067)	(0.065)	(0.184)	(0.190)
Dynamic Missing	-0.027	-0.074**	-0.070*	0.153***	-0.085	0.778***
	(0.020)	(0.032)	(0.038)	(0.054)	(0.176)	(0.188)
Long-Term Missing	-0.037	-0.042	-0.042	1.271***	3.195***	5.176***
	(0.026)	(0.040)	(0.046)	(0.072)	(0.233)	(0.272)
Time FEs	Yes	Yes	Yes	Yes	Yes	Yes
Fund FEs	Yes	Yes	Yes	Yes	Yes	Yes

*p<0.1; **p<0.05; ***p<0.01

Note: The classification of missing months follows the approach described in section 2. Values are winsorized at the 1st and 99th percentile. Standard errors are clustered at the fund and month-level. Data from CRSP for the US and Lipper for Europe.

4.2 Revisiting Empirical (In)Consistency

As in Schwarz and Potter (2016), I investigate fund manager skill due to its prominence in the literature. I further look at socially responsible investing and portfolio liquidity management because of the potential influence of strategic usage of portfolio reporting.

This section extends prior analyses in two ways. First, it investigates funds not subject to the exhaustive portfolio disclosure found in the US, and for which data coverage is poorer. Second, it decomposes the missingness into the patterns studied in the previous sections, thus shedding light on why certain empirical analyses could suffer from consequential biases.

4.2.1 Fund Manager Skill

The skill of fund managers is one of the most researched areas in the mutual fund literature. Numerous measures rely on portfolio data to decompose the different facets of skill, which can in turn be affected by the availability of such data. To evaluate this difference in skill measurement, I use the approach proposed by Kacperczyk et al. (2014), which distinguishes stock selection and market timing abilities by studying US equity funds from 1980 to 2005. The main finding of Kacperczyk et al. (2014) is that the most proficient investment managers skillfully adapt their strategies to reflect the evolving dynamics of economic cycles, with better stock-picking in expansions and stock-timing during recessions. This analysis could be particularly sensitive to gaps in missing data as the timing of a change in position cannot be assessed accurately. Since mandatory portfolio disclosure was limited to two semi-annual reports prior to 2004, 49% the funds in this sample disclosed quarterly portfolios and virtually none reported every month.

To replicate the baseline analysis, I obtain fund portfolio data from OPC, the portfolio reporting schedule and fund-level from controls from Lipper IM, and identify the monthly returns of individual assets from Datastream. Following Kacperczyk et al. (2014), I drop funds with total net asset below 5 million euros, with fewer than 10 unique stock holdings or with less than 80% of their holdings invested in equity. The final sample consists of 93,109 fund-months (of which two thirds are missing in Lipper), covering the investment behavior of 1,479 equity funds over the period 2012/01 to 2024/06. This covers three periods what the Saint Louis Federal Reserve Bank's OECD Europe Recession Indicator identifies as recessions: 2011/06 to 2013/02, 2015/09 to 2016/07 and 2018/01 to 2020/05.

The regressions on the complete dataset are roughly in line with the paper's findings. The top quartile of stock-picking funds mostly outperform other funds through stock-picking in expansions, and do not underperform in timing during recessions. This contrasts with what would be found if using only the commercially-available data[20]. In table 5 I find that the top funds instead exhibit

[20] Since the baseline analyses of Kacperczyk et al. (2014) seem to rely on up to 5-month forward imputation to obtain their monthly portfolio values, this table provides estimates for no estimates that are based on the complete dataset minus the fund-months missing at the intensive and dynamic margin. The other analyses in this section will provide this breakdown.

nearly no relative time-variation in investment skill deployment. The decrease in sample size therefore leads to an erroneous conclusion, which points to the limits of using portfolio data to study manager skill when this data is not consistently available.

Table 5: Measuring time-varying fund skill (2012/01 to 2024/06)

I test the baseline regression of Kacperczyk et al. (2014) on French funds, both on all fund portfolios and on only the available ones in Lipper. I find roughly consistent results when testing this on the complete dataset, as well as when dropping funds that never report: higher picking skills during expansions that are traded for relative increased timing-skill during recessions. When only using reported portfolios, however, I find no strategic switch.

	Dependent variable:					
	(1)	(2)	(3)	(4)	(5)	(6)
Timing	-0.261^{***}	-0.032	-0.216^{***}	-0.015	-0.306^{***}	-0.244^{***}
	(0.031)	(0.055)	(0.037)	(0.066)	(0.050)	(0.092)
Picking	0.419^{***}	-0.023	0.382^{***}	-0.004	0.604^{***}	0.146^{***}
	(0.029)	(0.038)	(0.035)	(0.046)	(0.049)	(0.062)
State	Expansion	Recession	Expansion	Recession	Expansion	Recession
Included funds	All	All	w/o Extensive	w/o Extensive	Available	Available
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	68,517	24,592	47,258	17,422	20,566	5,984
R^2	0.011	0.003	0.010	0.002	0.018	0.003
Note:						*p<0.1; **p<0.05; ***p<0.01

An important stream in the manager skill literature seeks to identify skill factors and predictors. Pástor et al. (2017) identifies past portfolio turnover as a positive predictor of fund performance, with stronger predictive power in illiquid assets. I test the baseline regression on French funds, both on all fund portfolios and on only the available ones in Lipper. I find roughly consistent results when testing this on the available French equity, mixed assets and bond fund portfolio data (Column 5 in table 6): higher turnover in the past year is associated with higher fund performance relative to its relevant benchmark for bond funds.

When including non-reported portfolios, however, the direction of this effect changes. By repeating this analysis without portfolio data that are missing in the three missingness patterns, it becomes clear that the dynamic margin plays the driving role: without these missing portfolios, the empirical estimates are virtually identical to those obtained from the commercial dataset for the latter two asset classes. The most probable explanation for this is that it is the cause of most missing portfolios here. Although many funds are missing completely, many are small enough or sufficiently concentrated to be dropped by the fund filters used in Pástor et al. (2017).

This insight affects investors who rely on performance metrics to identify skilled funds, as the inclusion of non-reported portfolios reveals a nuanced picture of fund turnover's impact on performance. Without addressing missing portfolio data, investors risk inaccurately forecasting fund performance and making suboptimal investment decisions.

Table 6: Portfolio turnover as a predictor of fund performance (2012/09 to 2021/12)

I test the baseline regression of Pástor et al. (2017) on French funds, both on all fund portfolios and on only the available ones in Lipper. I find roughly consistent results when testing this on the available French equity, mixed assets and bond fund portfolio data (Column 5): higher turnover in the past year is associated with higher fund performance relative to its relevant benchmark for bond funds. When including non-reported portfolios, however, the direction of this effect changes. This difference stems mostly from gaps in otherwise reporting funds (i.e., dynamic margin).

	Dependent variable:				
	Net Returns				
	(1)	(2)	(3)	(4)	(5)
Turnover (equity funds)	−0.030*	−0.024	−0.028	−0.020	−0.003
	(0.018)	(0.018)	(0.018)	(0.027)	(0.026)
Observations	54,473	52,721	52,194	28,367	24,336
R^2	0.021	0.022	0.022	0.028	0.035
	(1)	(2)	(3)	(4)	(5)
Turnover (mixed funds)	−0.024**	−0.023**	−0.028**	−0.014	−0.014
	(0.011)	(0.011)	(0.011)	(0.014)	(0.016)
Observations	39,820	37,998	37,629	20,892	16,879
R^2	0.029	0.030	0.028	0.042	0.044
	(1)	(2)	(3)	(4)	(5)
Turnover (bond funds)	0.014	0.016	0.014	0.042**	0.050**
	(0.011)	(0.012)	(0.011)	(0.020)	(0.024)
Included funds	All	w/o Extensive	w/o Intensive	w/o Dynamic	Available
Fund FEs	Yes	Yes	Yes	Yes	Yes
Time FEs	Yes	Yes	Yes	Yes	Yes
Observations	19,181	18,516	19,181	10,423	9,090
R^2	0.022	0.021	0.022	0.049	0.049
Note:					*p<0.1; **p<0.05; ***p<0.01

In addition to the portfolio turnover, the deviation in stock selection is a well-studied characteristic of manager skill. Cremers and Petajisto (2009) identifies the activeness of the investment strategy of funds as a simple, portfolio-based predictor of its relative net performance. To test this, they propose the now well-known Active Share metric, which measures the deviation by a fund of its benchmark. I test this on French equity funds, both on all fund portfolios and on only the available ones in Lipper, from 2012 to 2022.

This analysis reveals that using commercial and regulatory data can yield divergent results. Regulatory data produces a positive but statistically insignificant estimate, whereas the widely accessible commercial data results in a significant negative estimate. This discrepancy is primarily due to non-reporting funds, despite accounting for less than a third of missingness. The findings of the previous section helps explain this difference: fund portfolio concentration is a large and important predictor of reporting at the extensive margin.

Table 7: Active Share as a predictor of fund performance (2012/01 to 2022/12)

I test the baseline regression of Cremers and Petajisto (2009) on French equity funds, both on all fund portfolios and on only the available ones in Lipper, from 2012 to 2022. I find that relying on commercial and regulatory data can lead to different results. Whilst the regulatory data leads to an insignificant and positive estimate, the commonly available data leads to a significant negative one.

	Dependent variable:				
	Net Returns				
	(1)	(2)	(3)	(4)	(5)
Active Share	0.034	−0.053	0.047	0.017	−0.145**
	(0.040)	(0.046)	(0.041)	(0.049)	(0.062)
Included funds	All	w/o Extensive	w/o Intensive	w/o Dynamic	Available
Time FEs	Yes	Yes	Yes	Yes	Yes
Observations	46,927	38,523	44,884	28,886	18,439
R^2	0.0001	0.002	0.0004	0.003	0.012
Note:					*p<0.1; **p<0.05; ***p<0.01

4.2.2 Socially Responsible Investment

Socially Responsible Investment (SRI) has become the focus of a significant body of empirical research in the fund literature[21]. Early research on SRI funds focused on their performance relative to conventional funds (Bauer et al., 2005; Gil-Bazo et al., 2010; Nofsinger and Varma, 2014; Revelli and Viviani, 2015; Hartzmark and Sussman, 2019), but SRI fund managers may have an additional incentive to cultivate a robust ESG image to draw capital from investors with a heightened sensitivity to ESG standards (Bollen, 2007; Riedl and Smeets, 2017; Ammann et al., 2019; Hartzmark and Sussman, 2019; Quang and Nefzi, 2023; Spaans et al., 2024).

This leaves a potential for portfolio manipulation by SRI funds to provide an improved ESG image to investors. Kempf and Osthoff (2008) and F.Muñoz et al. (2022) address this by testing for semi-annual and quarterly window-dressing respectively, but use public and commercially-available portfolio information respectively[22]. These might not be representative of holdings on non-reported dates, either because of strategic non-reporting or the prevalence of missing portfolio data, differences that can be particularly strong in jurisdictions without mandatory portfolio disclosure. I therefore revisit this question by including all monthly portfolios, including those that are neither public nor provided to commercial databases.

Using the portfolio holdings, I compute three ESG scores for each fund-month: the mean carbon score, the share of the portfolio invested in brown securities[23], and the share invested in ESG-incompatible industries[24]. I use the following specification:

$$ESGScore_{i,t} = \beta_0 * SRI_i + \sum_j \beta_j * Controls_{i,j,t} + \alpha_t + \epsilon_{i,t} \tag{4}$$

with SRI_i a dummy identifying SRI funds. Controls include the logged fund size, logged number of unique holdings, portfolio concentration (HHI) and the share of the portfolio covered by ESG data. I obtain fund portfolio data from OPC, the SRI label from Lipper IM and the carbon scores are computed from the company and country-level climate scores from the data provider ISS. I focus on equity funds with assets under management of at least 5 million euros, and further drop fund portfolios of which less than 65% is allocated a Carbon Score or Industry. To maintain the relevance and accuracy of the analysis, I limit the study period to two years before and after the validity of the carbon score values, as these scores remain time-invariant within the dataset. The final sample consists of 17,844 fund-months (of which half are missing in Lipper), covering 499 equity funds of which 171 have an SRI label[25]. Standard errors are clustered at the month-level. Table 8) summarizes the resulting estimates.

[21] Fabregat-Aibar et al. (2019) identifies 209 papers focused on SRI funds published between 1988 and 2018.

[22] This information is easily accessible on platforms often used by investors, such as Lipper IM and Morningstar Direct, and, when not required by mandatory reporting, only includes funds that voluntarily provide this data.

[23] Defined as those with a Carbon Score lower than 50 (a threshold established by ISS).

[24] Those industries are based on the GICS classification: Metals & Mining, Oil, Gas & Consumable Fuels, Aerospace & Defense, Tobacco, Airlines, Gas Utilities.

[25] 25 of the SRI funds and 74 of the non-SRI funds never reported a portfolio.

Overall, I find that SRI funds invest, in fact, more socially responsibly than other equity funds. SRI fund portfolios hold less brown and ESG-incompatible securities, and have a higher weighted mean carbon score. Estimates with and without non-reported fund-months are very similar, despite sizable gaps in reporting. This indicates that SRI funds do not engage in window-dressing by deliberately withholding portfolios with varying ESG scores. Consequently, it is unlikely that earlier studies have been impacted by a missing data bias stemming from this practice.

Table 8: ESG scores in reported and non-reported French equity fund portfolios (2017/12 to 2021/12)

	Dependent variable:				
	Share of Brown holdings (%)				
	(1)	(2)	(3)	(4)	(5)
SRI	-3.367^{***}	-3.115^{***}	-3.356^{***}	-3.636^{***}	-3.507^{***}
	(0.061)	(0.087)	(0.066)	(0.131)	(0.144)
Observations	22,319	21,186	21,154	13,863	11,565
R^2	0.161	0.184	0.154	0.162	0.180
	Mean Portfolio Carbon Score				
	(1)	(2)	(3)	(4)	(5)
SRI	1.787^{***}	1.763^{***}	1.748^{***}	1.917^{***}	1.925^{***}
	(0.043)	(0.043)	(0.043)	(0.048)	(0.049)
Observations	22,319	21,186	21,154	13,863	11,565
R^2	0.153	0.162	0.147	0.159	0.163
	Share of ESG-incompatible holdings (%)				
	(1)	(2)	(3)	(4)	(5)
SRI	-1.672^{***}	-1.626^{***}	-1.801^{***}	-1.631^{***}	-1.908^{***}
	(0.059)	(0.048)	(0.058)	(0.091)	(0.087)
Included funds	All	w/o Extensive	w/o Intensive	w/o Dynamic	Available
Controls	Yes	Yes	Yes	Yes	Yes
Time FEs	Yes	Yes	Yes	Yes	Yes
Observations	22,308	21,175	21,143	13,863	11,565
R^2	0.055	0.033	0.055	0.083	0.041
Note:					*p<0.1; **p<0.05; ***p<0.01

4.2.3 Portfolio Liquidity Management

There is no academic consensus on how funds manage their liquidity. Whereas some studies describe a clear pecking-order of liquidation in which fund managers first sell their more liquid assets to meet redemptions - sometimes called a "waterfall" approach (Chernenko and Sunderam, 2016; Zeng, 2017; Choi et al., 2020), others find evidence that fund managers preserve the liquidity of their portfolio to prepare for future outflows (Morris et al., 2017). Jiang et al. (2021), in contrast, describe the use by fund managers of dynamic strategies, where funds mainly rely on liquid assets to accommodate flows under quiet market circumstances but proportionally scale down their holdings to preserve portfolio liquidity during times of stress.

Thus, a natural question is whether the empirical conclusions derived from the commercial data are consistent with estimates that are free from missing data bias. To evaluate fund liquidity management, I emulate the approach of Jiang et al. (2021). In line with this approach, I restrict the analysis to quarterly portfolios of funds identified as bond funds, with fixed income portfolios consisting of at least 50 percent of corporate debt. The estimated panel regression relates quarterly changes in fund holdings across asset classes to contemporaneous fund flows:

$$\Delta CashShare(\%)_{i,t} = \beta_0 + \beta_1 * NetInflow_{i,t} + \beta_2 * NetOutflow_{i,t} + \sum_j \beta_j * Controls_{i,j,t} + \alpha_i + \alpha_t + \epsilon_f \tag{5}$$

with $\Delta CashShare(\%)_{i,t}$ the change in the fraction of fund assets invested in cash by fund i from period t-1 to period t. $NetInflow_{i,t}$ ($NetOutflow_{i,t}$) equals net fund flows, in decimal, for fund i during quarter t if it is above (below) 0, and 0 otherwise, such that β_1 and β_2 capture the effects of net investor purchases and redemptions on changes in fund asset allocation. Controls include the log of fund age, the natural logarithm of size, and fund turnover. I obtain fund portfolio data from OPC and fund returns and flows from Lipper IM. The final sample consists of 7,527 fund-quarters (of which half are missing in Lipper), covering the liquidity management of 395 bond funds over the period 2011/09 to 2021/12[26]. Standard errors are clustered at the fund and date-level. Table 9) summarizes the resulting estimates.

I find that the resulting estimates differ between the reported and non-reported data. Most notably, the fund-quarters with reported portfolios appear to rely significantly less on cash to meet redemptions than their missing counterparts. A one percent net outflow is associated with a 2.05 percent reduction of cash in the fund's portfolio in the commercial data, which is three times less than for fund-quarters in the missing data.

Most missing fund portfolios stem from inconsistent reporting, and the estimates from the three pattern-specific regressions reflect this. The findings of the previous section suggest that the role of large flows on the probability of portfolio reporting (see Table 3) could explain this. Relying on the

[26]This sample is smaller, but of comparable size to the 578 bond funds and 13,868 fund-quarters in the sample of Jiang et al. (2021).

commercial data alone would (erroneously) suggest that French funds adopt a more balanced liquidity management strategy than would be found without missing data bias. This misrepresentation is critical because it could lead investors and policymakers to misinterpret the fund's resilience to liquidity events and its potential reactions during turbulent times with significant outflows.

Table 9: Liquidity Management of Corporate Bond Funds for available and missing fund-quarters

	Dependent variable:				
	$\Delta CashShare(\%)$				
	(1)	(2)	(3)	(4)	(5)
Net Inflows	1.727***	1.786***	1.728***	1.834***	2.004***
	(0.388)	(0.415)	(0.389)	(0.455)	(0.420)
Net Outflows	4.041***	3.903***	4.112***	2.535***	2.052**
	(0.750)	(0.792)	(0.756)	(0.783)	(0.774)
Included funds	All	w/o Extensive	w/o Intensive	w/o Dynamic	Available
Controls	Yes	Yes	Yes	Yes	Yes
Fund FEs	Yes	Yes	Yes	Yes	Yes
Time FEs	Yes	Yes	Yes	Yes	Yes
Observations	7,527	6,758	7,514	4,755	3,973
R^2	0.046	0.046	0.046	0.071	0.076
Note:					*p<0.1; **p<0.05; ***p<0.01

In summary, these analyses reveal that the omission of missing fund portfolios and non-reporting funds can lead to significantly different conclusions compared to those derived from a complete dataset. The estimates of the determinants of fund performance exhibit sensitivity to this missingness, with reliance on commercial data alone potentially leading to contrary findings. Similarly, estimates regarding the liquidity management of bond funds are affected, as fund flows differ markedly during months when a fund did not report a portfolio. However, this issue is not universal. The analysis indicates that equity SRI funds are significantly more socially responsible/green than their non-SRI counterparts, even within the context of missing data, suggesting that they adhere to their stated principles. Overall, these findings underscore the importance of accounting for missing data in empirical analyses involving fund portfolio data.

5 Solutions to Missingness?

In the previous section, I examined the impact of missing data on empirical research, illustrating how the distribution of missing portfolio data gaps can skew results and undermine the validity of analyses. This section delves into three primary approaches for mitigating the challenges posed by missing data: 1) merging multiple databases to augment the data coverage, 2) imposing fund filters to drop the most problematic forms of data missingness, and 3) applying imputation techniques to fill in data gaps.

5.1 Merging Multiple Databases

A first solution to missing data is to merge multiple commercial databases. Since gaps in reporting largely originate at the fund company-level, this can take advantage of the tendency of certain fund management companies to report more to one commercial data provider than the other. Table 10 decomposes the coverage by types of missingness and shows where the merged dataset would improve coverage relative to starting with only one of the two databases, and figure 9 further displays the evolution of both the common and unique coverage of the two datasets in Appendix B.

I find that between January 2010 and December 2024, merging the commonly-used Lipper/CRSP and Morningstar databases achieves major increases in portfolio coverage relative to only using Lipper/CRSP, but only marginal increases relative to only using Morningstar. In Europe, adding Morningstar to Lipper increases the coverage by 22.63 percentage points, whilst the opposite increases coverage by 4.14 percentage points. In the U.S., Morningstar again dominates: combining it with CRSP improves coverage by 5.89 percentage points, while adding CRSP to Morningstar yields almost no benefit (0.05 percentage points).
In the US, adding adding Morningstar to CRSP increases the coverage by only 5.89 percentage points, whilst the opposite increases it by only 0.05 percentage points. What matters, however, is whether these increases are substantial enough to limit the missing data bias.

In the U.S., Morningstar's relatively high standalone coverage means that merging it with CRSP meaningfully reduces the overall share of missingness in the latter. Although intensive-margin gaps remain nearly unchanged (declining only slightly from 24.11% to 24.06% of observations), the more problematic dynamic and long-term gaps become roughly three times less frequent. These residual gaps may still affect empirical results, but their impact is likely to be far smaller.

Table 10: Portfolio Data and categorization of portfolio data missingness in CRSP and Lipper

(a) Morningstar + CRSP (US), 2010/01 to 2024/12[27]

Availability/missingness	**Observations**	**Share (%)**	**Lipper gain (%)**	**Morningstar gain (%)**
Portfolio available	1,123,327	73.04	(+5.89)	(+0.05)
Missing:	414,643	26.96	(-5.89)	(-0.05)
Intensive	369,982	24.06	(-0.05)	(+0.02)
Extensive	513	0.03	(-0.01)	(-0.00)
Dynamic (short-term)	14,681	0.95	(-4.02)	(-0.02)
Dynamic (long-term)	29,226	1.92	(-1.84)	(-0.05)

(b) Morningstar + Lipper (Europe), 2010/01 to 2024/12

Availability/missingness	**Observations**	**Share (%)**	**Lipper gain (%)**	**Morningstar gain (%)**
Portfolio available	3,877,722	61.40	(+22.63)	(+4.14)
Missing:	2,437,746	38.60	(-22.63)	(-4.14)
Intensive	421,123	6.67	(+0.61)	(-0.16)
Extensive	698,812	11.07	(-8.64)	(-6.16)
Dynamic (short-term)	231,643	3.67	(-4.37)	(+0.08)
Dynamic (long-term)	1,086,168	17.20	(-10.22)	(+2.12)

For European funds, however, the benefits of merging commercial databases are limited. This may appear counterintuitive, as overall coverage increases substantially. However, most of the improvement comes from funds that report to only one of the two providers, reducing the share of permanently missing funds (the extensive margin). This comes at the cost of a higher share of long-term gaps, since many of these newly covered funds did not report from inception. Consequently, consistent with the findings of the previous section, the forms of missingness most likely to bias empirical analysis remain largely unchanged or are even exacerbated.[28]

In conclusion, merging CRSP with Morningstar (or simply relying on Morningstar directly) is likely sufficient to eliminate most of the missing portfolio data bias in analyses on US funds, but unlikely to be sufficient on analyses performed on European funds.

5.2 Fund filters

An alternative strategy for addressing missing portfolio data is to apply filters that limit the sample to fund-months where missingness is less likely to be systematically related to other characteristics. While such filtering may be undesirable for certain types of analysis, it can effectively reduce the magnitude of bias arising from non-random data absence in many contexts.

[27] U.S. Morningstar figures include only those funds that could be matched to CRSP. Because some Morningstar funds could not be matched, the total number of fund–months reported here is smaller than Morningstar's full universe. To ensure a like-for-like comparison with CRSP and CRSP + Morningstar, percentage changes in coverage are computed only for the matched subset common to both databases.

[28] A further hurdle to merging datasets is cost: acquiring multiple commercial databases can be prohibitively expensive for many - if not most - researchers.

One possible approach involves relying exclusively on portfolio disclosures mandated by regulation. In the United States, for example, mutual funds submit quarterly portfolio filings.[29] Although these mandatory disclosures are comprehensive and systematic, their lower frequency may limit their usefulness in studies requiring higher-frequency data. As shown by Elton et al. (2010), relying on quarterly data can lead to materially different conclusions in such cases.

A second approach is to restrict the analysis to fund-months in which portfolio data is more reliably available. Consistent with the findings in the previous section, the most problematic instances of missing data tend to occur in the early life of a fund—particularly during its initial months. Table 11 presents the impact of excluding the first six months of each fund's existence on the observed differences between fund-months with and without portfolio data. Although this exclusion removes only a modest share of observations (approximately 3.5 percent of fund-months), it substantially reduces the observed bias in flow measures, both contemporaneous and those from the previous and subsequent quarters.

In summary, omitting a fund's first few months from the analysis is a relatively blunt but effective method for mitigating biases introduced by missing portfolio data.

Table 11: Differences in returns and flows for fund-months with and without available portfolio data, in the US and Europe, by type of data absence

CRSP for US funds (2010/01 to 2024/12)

	Return	Past Return	Next Return	Flows	Past Flows	Next Flows
Intensive Missing	0.029**	-0.047**	0.007	-0.019	0.376**	-0.068
	(0.014)	(0.022)	(0.021)	(0.075)	(0.156)	(0.154)
Dynamic Missing	0.002	-0.021	-0.093**	0.230***	0.681***	0.629***
	(0.028)	(0.051)	(0.045)	(0.071)	(0.198)	(0.164)
Long-Term Missing	-0.019	-0.021	0.084	3.610***	5.261***	16.031***
	(0.058)	(0.092)	(0.099)	(0.260)	(0.452)	(1.054)
Intensive Missing Filt	0.029**	-0.046**	0.007	0.030	0.291**	0.138
	(0.014)	(0.022)	(0.022)	(0.071)	(0.145)	(0.148)
Dynamic Missing Filt	0.003	-0.021	-0.091**	0.196***	0.606***	0.677***
	(0.028)	(0.051)	(0.046)	(0.069)	(0.186)	(0.151)
Long-Term Missing Filt	-0.012	-0.027	0.082	0.498***	2.767***	1.942***
	(0.063)	(0.091)	(0.120)	(0.106)	(0.318)	(0.283)

Lipper for European funds (2010/01 to 2023/12)

[29] Since 2004, the SEC has required mutual funds to file Forms N-CSR and N-Q quarterly. In 2016, these were replaced by Form N-PORT, which mandates monthly reporting, though only quarter-end filings are made public.

	Return	Past Return	Next Return	Flows	Past Flows	Next Flows
Intensive Missing	-0.051	-0.080	-0.056	-0.162**	-0.207	-0.407**
	(0.048)	(0.074)	(0.067)	(0.065)	(0.184)	(0.190)
Dynamic Missing	-0.027	-0.074**	-0.070*	0.153***	-0.085	0.778***
	(0.020)	(0.032)	(0.038)	(0.054)	(0.176)	(0.188)
Long-Term Missing	-0.037	-0.042	-0.042	1.271***	3.195***	5.176***
	(0.026)	(0.040)	(0.046)	(0.072)	(0.233)	(0.272)
Intensive Missing Filt	-0.049	-0.031	-0.013	-0.069	-0.051	0.099
	(0.042)	(0.062)	(0.061)	(0.065)	(0.174)	(0.163)
Dynamic Missing Filt	-0.013	-0.030	-0.015	0.040	-0.062	0.325***
	(0.012)	(0.022)	(0.023)	(0.034)	(0.113)	(0.105)
Long-Term Missing Filt	-0.015	-0.030	0.012	0.338***	1.503***	1.344***
	(0.020)	(0.031)	(0.037)	(0.040)	(0.144)	(0.134)

*p<0.1; **p<0.05; ***p<0.01

Note: The classification of missing months follows the approach described in section 2. Values are winsorized at the 1st and 99th percentile. Standard errors are clustered at the fund and month-level. Data from CRSP for the US and Lipper for Europe.

5.3 Imputing Portfolio Data

The final approach consists in imputing portfolio data, based on available ones. Recent research introduced techniques to derive individual, daily trades from commonly-available data (Farrell, 2018; Bongaerts et al., 2024). These imputation methods reconstruct the intervening daily portfolios that best reconcile a fund's reported daily returns with the observed returns of its constituent securities[30]. However, the bias introduced by missing portfolio information is most acute precisely in the situations where neither a preceding nor a subsequent portfolio disclosure exists. As documented in Tables 4 and 11, funds-months with no available or adjacent portfolio exhibit markedly larger distortions in both contemporaneous and adjacent-period flow measures. Techniques that interpolate the composition of a fund between two observed portfolios therefore cannot be applied in these cases.

Wen no adjacent portfolio is available, the only practical (and therefore widely used) alternative is simple **forward** or **backward imputation**. Forward imputation uses the last observed portfolio to fill subsequent gaps, whereas backward imputation projects the next available portfolio backward to earlier dates. This method is simple to implement, but its effectiveness depends on portfolio

[30]Their effectiveness, however, is fund-specific. Accuracy is highest when performance is driven primarily by listed securities and cash—typical of conventional equity funds—whose daily price movements are fully observable. By contrast, precision diminishes when portfolios contain substantial exposures to assets whose day-to-day valuations are unobservable (e.g., real estate), when funds rely heavily on complex derivatives, or when returns are distorted by stale pricing. The latter is common in bond portfolios, where infrequent price updates break the link between reported fund returns and the contemporaneous returns of the underlying securities, thereby reducing the reliability of the imputed holdings.

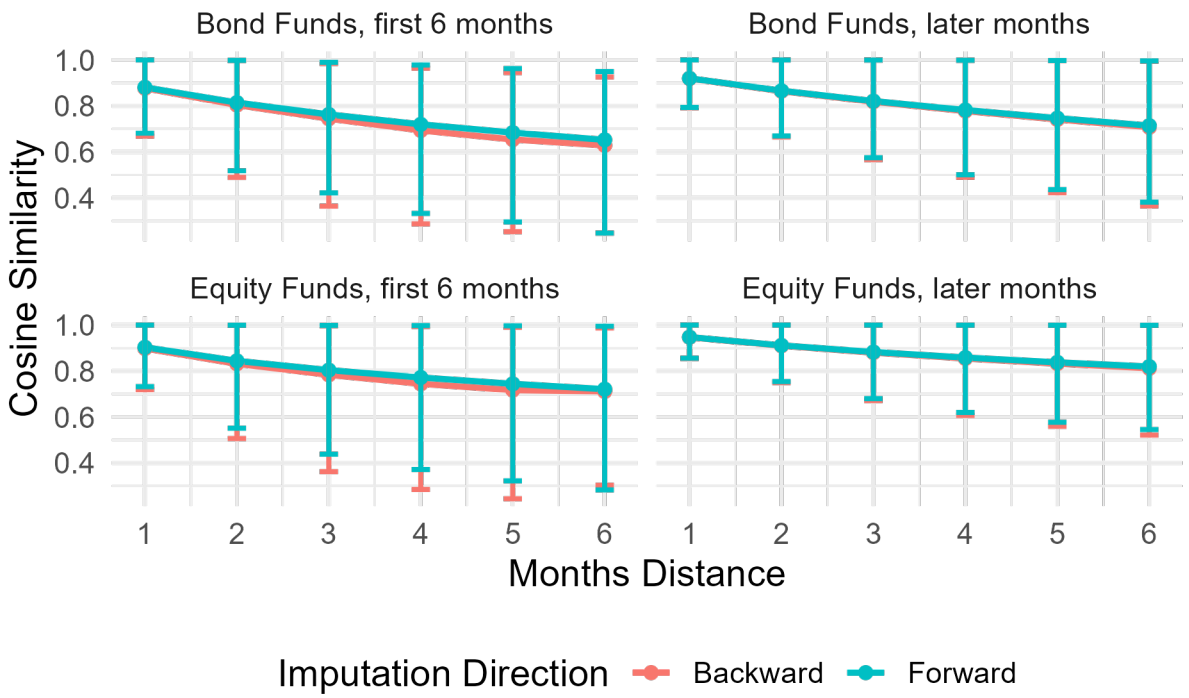
stability—which can only be evaluated when actual portfolio data are available. To assess the accuracy of both forward and backward imputation across varying time gaps, I again rely on the Banque de France's comprehensive OPC dataset, which provides complete portfolio histories for a wide range of funds.

Figure 4 presents the resulting distribution of cosine similarity between imputed and actual portfolios, computed across gaps ranging from one to six months. As expected, the accuracy of imputation declines in a convex fashion as the gap widens. Forward and backward imputation perform similarly, with accuracy consistently higher for equity funds than for bond funds—likely due to the greater tendency of bond funds to adjust their holdings more frequently and alter the set of securities in which they invest. The figure also reveals that imputation performance is substantially lower for younger funds, particularly beyond one-month gaps, whereas older funds maintain a higher degree of stability, with two-month gaps showing comparable performance.

To assess the incremental value of imputation schemes that exploit both a preceding and a subsequent disclosure, I implement the return-based reconstruction algorithm of Bongaerts et al. (2024) to fill two- and five-month gaps for actively managed French equity funds. Appendix B shows that this informed method does outperform simple forward imputation for traditional equity portfolios, delivering higher cosine similarity and thus a closer match to the actual holdings. The improvement, however, is quantitatively modest—particularly for the two-month gap—and comes at a substantially higher computational cost. Its principal advantage lies in the daily granularity it provides: by reconstructing a full sequence of intervening portfolios, the method is well suited to identifying individual trades and assessing trade timing, tasks for which single-side imputation is inadequate.

In conclusion, forward/backward imputation can be used to impute short-term (1 to 2 month gaps) gaps, but display lower performance for younger funds.

Figure 4: Portfolio Similarity Across Fund Lifecycles and Asset Classes, 2011/07 to 2023/12



This figure shows the average cosine similarity between each imputed portfolio and the corresponding actual (observed) portfolio. The x-axis indicates the number of months between the portfolio used for imputation and the target portfolio (e.g., a gap of 3 means weights from t3 are used to impute the portfolio at time t). Results are shown separately for equity and bond funds, and by fund age (first six months vs. later months). Forward and backward imputation directions are distinguished to highlight differences in portfolio stability over time. Error bars depict the distribution's 10th and 90th percentiles. Data from OPC. Author's calculations.

6 Conclusion

Missing portfolio disclosures continue to bias fund-level empirical work. I offer a classification and stylized facts of the absence of data informed by the data collection procedure of commercial data providers, and find that different types of data absence have different determinants. Not all gaps are equally consequential. The most influential are dynamic and long-term gaps—periods in which no portfolio is reported in either the prior or the subsequent month. These gaps, concentrated in young funds and coinciding with large flows, differ from gaps created merely by lower reporting frequencies (e.g., monthly vs quarterly), which contribute little to the overall bias.

Three practical remedies are examined. First, database merging all but removes the problem for U.S. funds when CRSP is combined with Morningstar, yet a Lipper + Morningstar merge leaves sizable holes in European coverage. Second, age-based filtering - dropping each fund's first six months - provides a blunt but highly effective correction, sharply reducing bias with only a modest loss of observations. Third, portfolio imputation remains indispensable in many analyses when coverage gaps persist. Simple forward or backward fills perform well for one- to two-month interruptions when only a single neighboring snapshot exists; more elaborate return-based interpolations add value chiefly when both adjacent portfolios are available and the gap exceeds two months.

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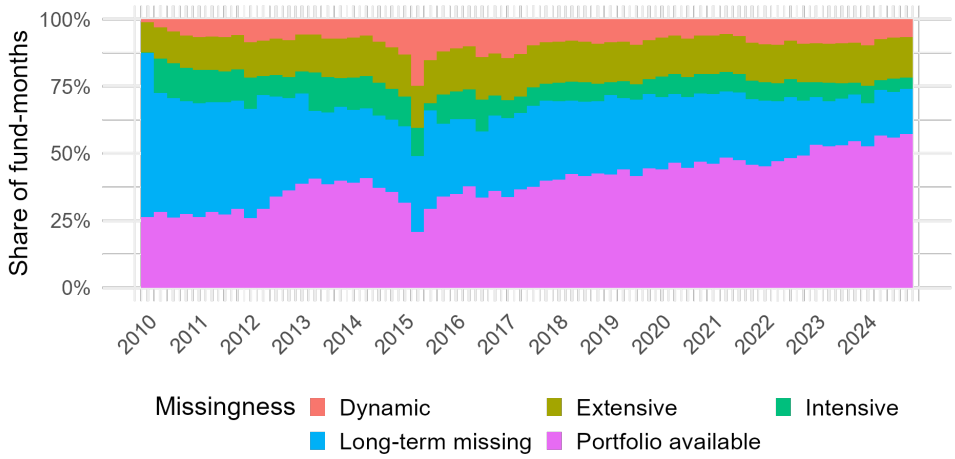
7 Appendix

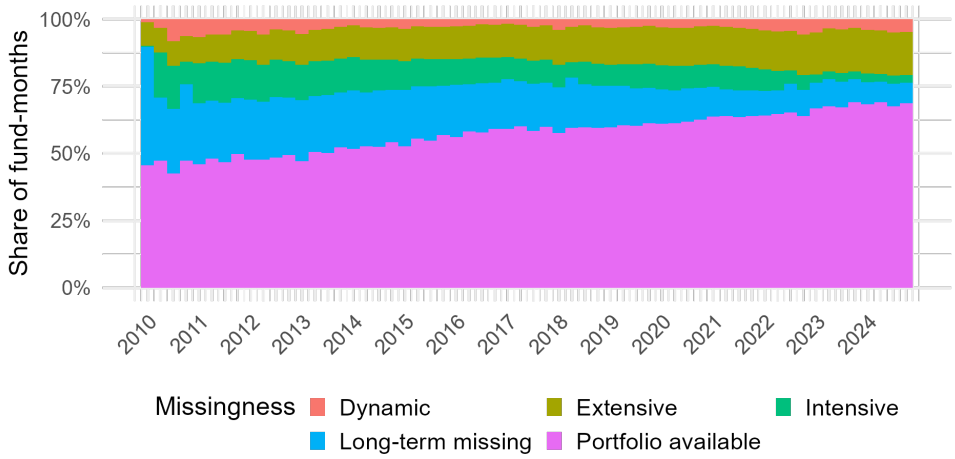
7.1 Appendix A: Portfolio Data reporting across Fund Domicile

Figure 5: Evolution portfolio availability of European funds in Lipper, 2010Q1 to 2024Q4



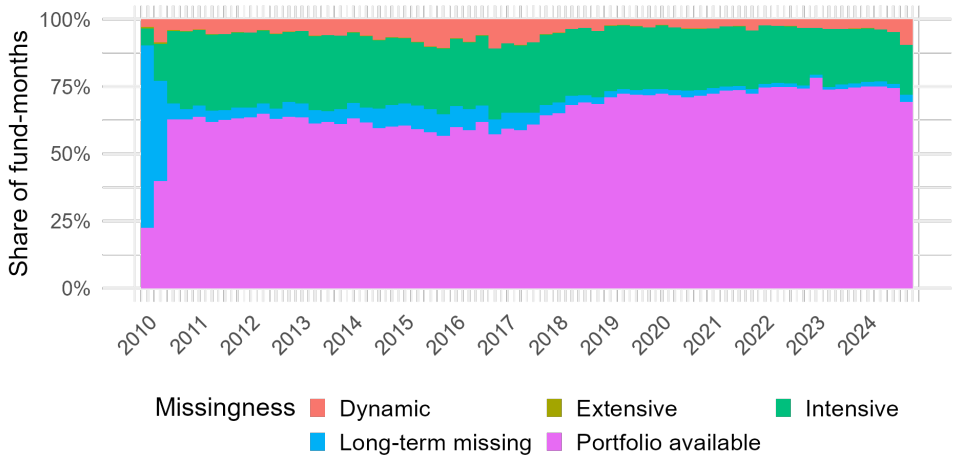
This graph displays the evolution of the reported data and the breakdown of missing data for European funds. Most absent portfolio data is missing because of permanent or long-term non-reporting. The large drop in 2015 stems from Lipper data being mostly unavailable for two months. The data are aggregated to a quarterly frequency for clarity. Data from Lipper IM.

Figure 6: Evolution portfolio availability of European funds in Morningstar, 2010Q1 to 2024Q4



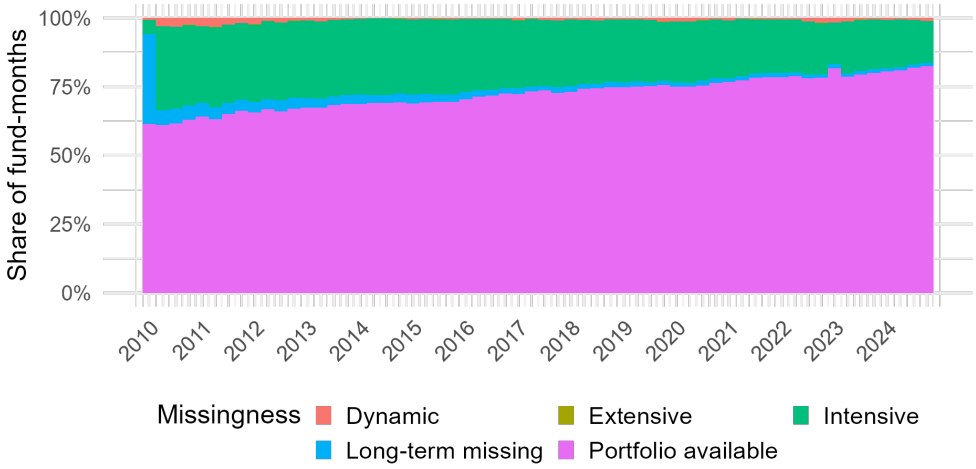
This graph displays the evolution of the reported data and the breakdown of missing data for European funds. Most absent portfolio data is missing because of permanent or long-term non-reporting. The data are aggregated to a quarterly frequency for clarity. Data from Morningstar Direct.

Figure 7: Evolution portfolio availability of US funds in CRSP, 2010Q1 to 2024Q4



This graph displays the evolution of the reported data and the breakdown of missing data for US funds. Most absent portfolio data is missing due to the lower reporting frequency (intensive margin) of certain funds, with other causes most prominent from 2014 to 2018. The data are aggregated to a quarterly frequency for clarity. Data from CRSP.

Figure 8: Evolution portfolio availability of European funds, 2010Q1 to 2024Q4



This graph displays the evolution of the reported data and the breakdown of missing data in Morningstar, for US funds. Most absent portfolio data is missing due to the lower reporting frequency (intensive margin) of certain funds. The data are aggregated to a quarterly frequency for clarity. Data from Morningstar Direct. Author's calculations.

7.2 Appendix B: Additional Graphs on the Performance of Considered Solutions

7.2.1 Appendix B.1: Imputing monthly portfolios using adjacent fund-months

I adapt the approach of Bongaerts et al. (2024) to impute the 2-month and 5-month gaps of 50 active French equity funds in the second semester of 2018 with a fixed 100 iterations, and compare the imputations with the actual end-of-month portfolios. I obtain portfolio data from OPC and daily security returns from Datastream. These funds have between 21 and 87 unique securities in their respective portfolios, and the return data covers on average 95% of their security holdings. I allow for two up to two changes from the original method. First, changing the produced output from the sum of daily-stock trades to the sum of weekly (or monthly) stock trades. The number of unknowns is thereby mechanically divided by the number of trading days per week (or month). Since gaps in reporting are often longer than the 2-month gaps stemming from quarterly reporting and can sometimes reach over a year in European data, this is a crucial simplification to allow computational feasibility. Second, changing the order in the lexicographic preferences, starting with the minimization of round-tripping across assets and then minimizing the return-error. This has the advantage to be less reliant on the return data, of which there is less following the reduction in imputation frequency. This reduces the risk of solutions with unrealistic trades that can stem from other drivers of daily fund returns (e.g., derivatives or even straightforward errors). I test three approaches that impute weekly approaches: model 1 reproduces the Bongaerts et al. (2024) approach at a weekly frequency, model 2 prioritizes the minimization of security holdings round-tripping over the return error, and model 3 prioritizes the minimization of cash and security holdings round-tripping over the return error.

Table 12 displays the distribution of the cosine similarities of the imputations and the actual portfolios. On the two-month gap, simple forward imputation performs relatively well, with Models 1 and 2 perform relatively similarly. Model 3 outperforms the others by about 1 percentage point at the median. On the five-month gaps, the differences are more pronounced, as Model 3 outperforms forward imputation by about 2 percentage points at the median. Notice that the relative performance of Model 3 is stronger towards the bottom of the distribution. It beats forward imputation by 1.6 and 0.3 percentage points at the first and third quartiles respectively for the 2-month gaps, and 2.3 and 1.1 percentage points for the 5-month gaps.

Table 12: Informed Imputation Performance (2018/06 to 2018/12)

This table displays the cosine similarities of the complete (including cash) imputed and actual portfolios of 50 French equity funds. Model 3 outperforms the other models on 2-month gaps and the 5-month gaps.

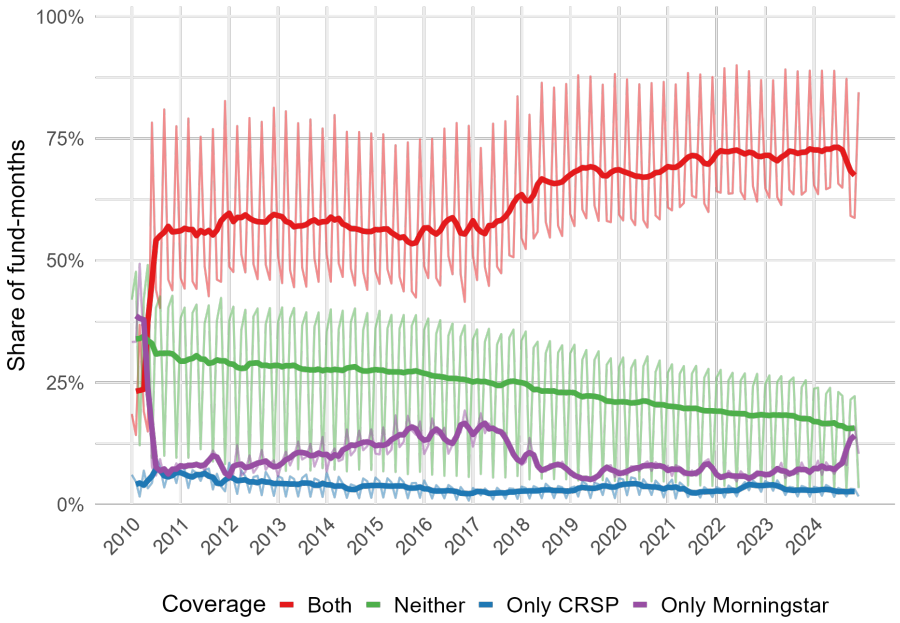
Approach	**Cosine Q1**	**Cosine Median**	**Cosine Mean**	**Cosine Q3**
Two-month imputations				
Forward	0.9216	0.9586	0.9396	0.9821
Model 1	0.9211	0.9616	0.9323	0.9805
Model 2	0.9218	0.9654	0.9353	0.9810
Model 3	**0.9372**	**0.9673**	**0.9482**	**0.9865**
Five-month imputations				
Forward	0.8766	0.9303	0.9173	0.9654
Model 1	0.8113	0.9067	0.8534	0.9651
Model 2	0.8246	0.9100	0.8529	0.9628
Model 3	**0.8992**	**0.9499**	**0.9308**	**0.9761**

7.2.2 Appendix B.2: Merging Lipper/CRSP with Morningstar

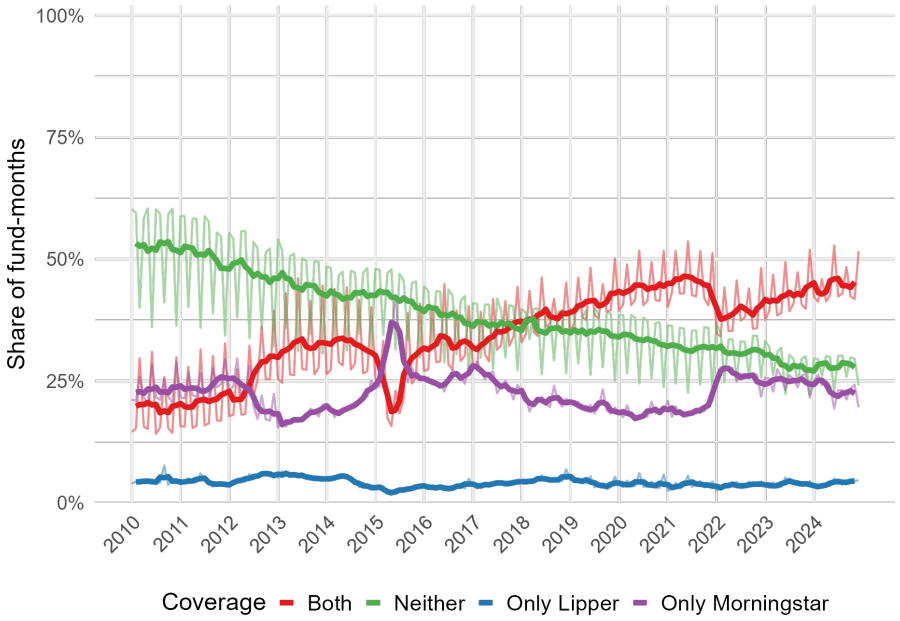
Figure 9: Portfolio Coverage of Lipper/CRSP and Morningstar for US and European funds, 2010 to 2024

These two graphs display the number of funds for which the datasets have a portfolio in a given month. The lines smooth quarter-end seasonality in portfolio coverage.

(a) United States



(b) Europe



7.3 Appendix C1: European regressions

Table 14: Drivers of the extensive and intensive margins of voluntary portfolio reporting in Europe, logit (feglm)

This table displays the panel logistic regressions on the extensive and intensive margins of reporting for European funds, from 2014 to 2024. Standard errors are clustered at the fund and month-level. I tackle the incidental parameter bias problem by applying the analytical bias correction derived by Fernández-Val and Weidner (2016). 38% of fund-months are of fund companies with homogeneous reporting frequencies at the extensive margin, against 69% at the intensive margin. Portfolio-level characteristics are significant predictors at the extensive margin.

	Dependent variable:					
	Extensive Reporting			Intensive Reporting		
	(1)	(2)	(3)	(4)	(5)	(6)
Log of Fund TNA	0.00	-0.01	-0.01	0.01	0.01	-0.04
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.02)
Log of Fund Age (in years)	0.42***	0.36***	0.53***	-0.08***	-0.10***	0.20***
	(0.02)	(0.02)	(0.03)	(0.02)	(0.02)	(0.04)
Alligned Reporting	-0.69***	-0.69***	-0.37**	-0.99***	-0.89***	-0.23
	(0.07)	(0.08)	(0.12)	(0.07)	(0.07)	(0.16)
Open Fund	1.81***	1.47***	1.95***	-0.47*	-0.59**	0.72
	(0.10)	(0.12)	(0.19)	(0.21)	(0.21)	(0.65)
Share Institutional	-0.50***	-0.59***	-0.15	1.56***	1.29***	0.21
	(0.07)	(0.06)	(0.10)	(0.06)	(0.06)	(0.14)
Competition (log)		-0.11***	-0.07**		-0.02	0.02
		(0.02)	(0.02)		(0.01)	(0.03)
Competition Reporting (ext)		6.22***	6.06***		-0.04	0.99
		(0.20)	(0.34)		(0.28)	(0.61)
Competition Reporting (int)		0.91***	1.19***		3.85***	0.00
		(0.19)	(0.30)		(0.16)	(0.44)
Bottom Quintile Alpha		-0.50***	-0.65***		-0.03	0.03
		(0.05)	(0.07)		(0.04)	(0.10)
Top Quintile Alpha		0.25***	0.12		0.22***	-0.03
		(0.07)	(0.08)		(0.05)	(0.10)
Time FEs	Yes	Yes	NA	Yes	Yes	NA
Fund Company-time FEs	No	No	Yes	No	No	Yes
Observations	1975992	1963207	1222683	1141111	1137583	356291
Deviance	1106265.87	994516.13	570176.78	1064494.90	1026011.41	189145.82
Pseudo R^2	0.06	0.14	0.31	0.07	0.10	0.49

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table 15: Drivers of the dynamic margin of voluntary portfolio reporting in Europe, logit (feglm)

This table displays the panel logistic regressions on the dynamic margin of reporting for European funds, from 2014 to 2024. Standard errors are clustered at the fund and month-level. I tackle the incidental parameter bias problem by applying the analytical bias correction derived by Fernández-Val and Weidner (2016). Dynamic gaps in reporting occur for most funds, but long-term gaps occur in a minority of funds (that in total form 20% of available fund-months).

	Dependent variable:					
	Dynamic (ST) Reporting			Dynamic (LT) Reporting		
	(1)	(2)	(3)	(4)	(5)	(6)
Log of Fund TNA	0.04^{***}	0.03^{***}	0.05^{***}	0.03^{***}	0.02^{***}	0.07^{***}
	(0.00)	(0.00)	(0.01)	(0.00)	(0.00)	(0.01)
Log of Fund Age (in years)	-0.01^{***}	0.15^{***}	0.17^{***}	0.25^{***}	1.70^{***}	1.88^{***}
	(0.00)	(0.01)	(0.01)	(0.00)	(0.01)	(0.02)
Alligned Reporting	-0.35^{***}			-0.75^{***}		
	(0.01)			(0.01)		
Open Fund	0.07^{**}			0.44^{***}		
	(0.02)			(0.02)		
Share Institutional	-0.16^{***}	0.03	-0.08^{*}	-0.04^{***}	-0.05	-0.01
	(0.01)	(0.02)	(0.04)	(0.01)	(0.03)	(0.05)
Competition (log)	-0.03^{***}	-0.14^{***}	-0.10^{*}	-0.07^{***}	-0.54^{***}	-1.43^{***}
	(0.00)	(0.03)	(0.05)	(0.00)	(0.04)	(0.08)
Competition Reporting (ext)	1.25^{***}	-0.24	0.49	3.65^{***}	4.17^{***}	3.21^{***}
	(0.03)	(0.16)	(0.25)	(0.03)	(0.18)	(0.34)
Competition Reporting (int)	0.60^{***}	0.56^{***}	0.09	2.28^{***}	2.83^{***}	2.48^{***}
	(0.03)	(0.09)	(0.15)	(0.02)	(0.09)	(0.17)
Return	0.00	0.00	0.00	0.01^{***}	-0.00	-0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Flow	0.00	0.00	-0.00	0.00^{***}	0.00	-0.00^{**}
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Time FEs	Yes	Yes	NA	Yes	Yes	NA
ISIN FEs	No	Yes	Yes	No	Yes	Yes
Fund Company-time FEs	No	No	Yes	No	No	Yes
Observations	1207105	1135472	857942	1456327	903047	686687
Deviance	993945.92	847646.41	406002.43	1611607.83	624749.52	246638.63
Pseudo R^2	0.049	0.166	0.476	0.095	0.498	0.733

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

7.4 Appendix C2: US regressions

Table 16: Drivers of the intensive margins of voluntary portfolio reporting in the US, logit

This table displays the panel logistic regressions on the extensive and intensive margins of reporting for US funds, from 2010 to 2024. Standard errors are clustered at the fund and month-level. I tackle the incidental parameter bias problem by applying the analytical bias correction derived by Fernández-Val and Weidner (2016). 42% of fund-months are of fund companies with homogeneous reporting frequencies at the intensive margin.

	Dependent variable:		
	Intensive Reporting		
	(1)	(2)	(3)
Log of Fund TNA	0.02*	-0.01	0.02
	(0.01)	(0.01)	(0.02)
Log of Fund Age (in years)	-0.07***	0.10***	0.10**
	(0.02)	(0.02)	(0.03)
Alligned Reporting	-0.46***	-0.59***	-0.29***
	(0.03)	(0.04)	(0.08)
Open Fund	0.34***	0.42***	0.41*
	(0.09)	(0.09)	(0.18)
Share Institutional		0.01***	0.01***
		(0.00)	(0.00)
Competition (log)		0.04	0.11**
		(0.02)	(0.04)
Competition Reporting (int)		2.11***	-0.10
		(0.60)	(1.09)
Bottom Quintile Alpha		0.02	0.02
		(0.04)	(0.07)
Top Quintile Alpha		0.06	0.01
		(0.04)	(0.07)
Time FEs	Yes	Yes	NA
Fund Company-time FEs	No	No	Yes
Observations	1056695	1056695	612696
Deviance	1396899.91	1359095.70	534350.94
Pseudo R^2	0.02	0.05	0.28

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Table 17: Drivers of the dynamic margin of voluntary portfolio reporting in the US, logit (feglm)

This table displays the panel logistic regressions on the dynamic margin of reporting for US funds, from 2010 to 2024. Standard errors are clustered at the fund and month-level. I tackle the incidental parameter bias problem by applying the analytical bias correction derived by Fernández-Val and Weidner (2016).

	Dependent variable:					
	Dynamic (ST) Reporting			Dynamic (LT) Reporting		
	(1)	(2)	(3)	(4)	(5)	(6)
Log of Fund TNA	0.05***	0.07***	0.10***	0.04***	-0.07***	-0.11***
	(0.00)	(0.01)	(0.01)	(0.00)	(0.01)	(0.03)
Log of Fund Age (in years)	0.10***	0.00	0.13***	0.49***	1.15***	1.65***
	(0.00)	(0.01)	(0.02)	(0.01)	(0.02)	(0.06)
Alligned Reporting	-0.08***			-0.19***		
	(0.01)			(0.01)		
Open Fund	0.41***			0.43***		
	(0.03)			(0.04)		
Share Institutional	0.00***	0.00	-0.00	0.01***	-0.00	-0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Competition (log)	0.20***	0.10	0.43**	0.14***	0.66***	1.10**
	(0.01)	(0.10)	(0.15)	(0.01)	(0.15)	(0.34)
Competition Reporting (int)	2.56***	2.74***	3.58***	7.24***	4.47***	8.70***
	(0.19)	(0.32)	(0.48)	(0.26)	(0.58)	(1.44)
Return	0.00	-0.00	-0.00	0.01***	-0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.01)
Flow	0.00***	0.00	-0.00	0.00***	0.00***	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Time FEs	Yes	Yes	NA	Yes	Yes	NA
ISIN FEs	No	Yes	Yes	No	Yes	Yes
Fund Company-time FEs	No	No	Yes	No	No	Yes
Observations	843270	708920	439403	813595	250154	98294
Deviance	411505.66	354784.86	188942.16	220154.89	102621.96	32494.29
Pseudo R^2	0.049	0.166	0.476	0.095	0.498	0.733

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$